32

Inductance

CHAPTER OUTLINE

32.1 Self-Induction and Inductance

32.2 *RL* Circuits

32.3 Energy in a Magnetic Field

32.4 Mutual Inductance

32.5 Oscillations in an *LC* Circuit

32.6 The *RLC* Circuit

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ32.1 **(i)** Answer (a). The mutual inductance of two loops in free space—that is, ignoring the use of cores—is a maximum if the loops are coaxial. In this way, the maximum flux of the primary loop will pass through the secondary loop, generating the largest possible emf given the changing magnetic field due to the first.

(ii) Answer (c). The mutual inductance is a minimum if the magnetic field of the first coil lies in the plane of the second coil, producing no flux through the area the second coil encloses.

OQ32.2 Answer (c). The fine wire has considerable resistance, so a few seconds is many time constants. The final current depends on the resistance of the wire, which has not changed; the current is not affected by the inductance of the coil because the current is not changing.

OQ32.3 Answer (b). The inductance of a solenoid is proportional to the number of turns squared, so cutting the number of turns in half makes the inductance four times smaller. Doubling the current would by itself make the stored energy $(\frac{1}{2}Li^2)$ four times larger, to just compensate.

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OQ32.4 The ranking is $\Delta V_L > \Delta V_{1\ 200\ \Omega} > 12.0\ \text{V} > \Delta V_{12\ \Omega}$. Just before the switch is thrown, the voltage across the 12-Ω resistor is very nearly 12 V (we assume the resistance of the inductor is small). Just after the switch is thrown, the current is nearly the same, maintained by the inductor, but this current is diverted through the 1 200-Ω resistor; thus, the voltage across the 1 200-Ω resistor is much more than 12 V, about 1 200 V, because the same current in the 12-Ω resistor now passes through a resistor 100 times as large. By Kirchhoff's loop rule, the voltage across the coil is larger still.

OQ32.5 Answer (d). The inductance of a solenoid is proportional to the number of turns squared (N^2), to the cross-sectional area (A), and to the reciprocal of the length of its axis (L). Coil A has twice as many turns with the same length of wire, so its circumference must be half as large as that of coil B: therefore, its radius is half as large and its area one quarter as large. For coil A the inductance will be different by the factor $N^2A/L \sim [2^2(1/4)]/2 = 1/2$.

OQ32.6 Answer (a). The energy stored in the magnetic field of an inductor is proportional to the square of the current. Doubling I makes $U_B = \frac{1}{2}Li^2$ get four times larger.

OQ32.7 Answer (d). The emf across an inductor is zero whenever the current is constant (unchanging), large or small.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ32.1 (a) We can think of Henry's discovery of self-inductance as fundamentally new. Before a certain school vacation at the Albany Academy about 1830, one could visualize the universe as consisting of only one thing, matter. All the forms of energy then known (kinetic, gravitational, elastic, internal, electrical) belonged to chunks of matter. But the energy that temporarily maintains a current in a coil after the battery is removed is not energy that belongs to any bit of matter. This energy is vastly larger than the kinetic energy of the drifting electrons in the wires. This energy belongs to the magnetic field around the coil. Beginning in 1830, Nature has forced us to admit that the universe consists of matter and also of fields, massless and invisible, known only by their effects.

The idea of a field was not due to Henry, but rather to Faraday, to whom Henry personally demonstrated self-induction. Still the thesis stated in the question has an important germ of truth.

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Henry precipitated a basic change if he did not cause it.

(b) A list today of what makes up the Universe might include quarks, electrons, muons, tauons, and neutrinos of matter; photons of electric and magnetic fields; W and Z particles; gluons; energy; charge; baryon number; three different lepton numbers; upness; downness; strangeness; charm; topness; and bottomness. Alternatively, the relativistic interconvertibility of mass and energy, and of electric and magnetic fields, can be used to make the list look shorter. Some might think of the conserved quantities energy, charge, . . . bottomness as properties of matter, rather than as things with their own existence.

CQ32.2 (a) The inductance of a coil is determined by (a) the geometry of the coil and (b) the "contents" of the coil. This is similar to the parameters that determine the capacitance of a capacitor and the resistance of a resistor. With an inductor, the most important factor in the geometry is the number of turns of wire, or turns per unit length. By the "contents" we refer to the material in which the inductor establishes a magnetic field, notably the magnetic properties of the core around which the wire is wrapped.

(b) No. The inductance of a coil is proportional to the flux through the coil per unit current, Φ/I, and the flux is proportional to the current I, so the inductance is independent of the current.

CQ32.3 When it is being opened. When the switch is initially standing open, there is no current in the circuit. Just after the switch is then closed, the inductor tends to maintain the zero-current condition, and there is very little chance of sparking. When the switch is standing closed, there is current in the circuit. When the switch is then opened, the current rapidly decreases. The induced emf is created in the inductor, and this emf tends to maintain the original current. Sparking occurs as the current bridges the air gap between the contacts of the switch.

CQ32.4 **(i)** (a) The bulb glows brightly right away, and then more and more faintly as the capacitor charges up. (b) The bulb gradually gets brighter and brighter, changing rapidly at first and then more and more slowly. (c) The bulb immediately becomes bright. (d) The bulb glows brightly right away, and then more and more faintly as the inductor starts carrying more and more current (the inductor eventually acts as a short).

(ii) (a) The bulb goes out immediately because current stops immediately (charge ceases to flow). (b) The bulb glows for a moment as a spark jumps across the switch. (c) The bulb stays

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lit for a while, gradually getting fainter and fainter as the capacitor discharges through the bulb. (d) The bulb suddenly glows brightly. Then its brightness decreases to zero, changing rapidly at first and then more and more slowly.

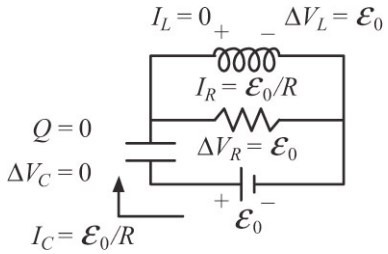
CQ32.5 (a) The coil has an inductance regardless of the nature of the current in the circuit. Inductance depends only on the coil geometry and its construction.

(b) Since the current is constant, the self-induced emf in the coil is zero, and the coil does not affect the steady-state current. (We assume the resistance of the coil is negligible.)

CQ32.6 (a) An object cannot exert a net force on itself. An object cannot create momentum out of nothing.

(b) A coil can induce an emf in itself. When it does so, the actual forces acting on charges in different parts of the loop add as vectors to zero. The term electromotive force does not refer to a force, but to a voltage.

CQ32.7 (a) The instant after the switch is closed, the capacitor acts as a closed switch, and the inductor acts to maintain zero current in itself. The situation is as shown in the circuit diagram of ANS. FIG. CQ32.7(a). The requested quantities are:

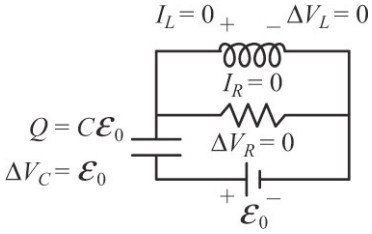


ANS. FIG. CQ32.7(a)

$$\boxed{I_L = 0,\ I_C = \frac{\mathcal{E}_0}{R},\ I_R = \frac{\mathcal{E}_0}{R}}$$

$$\boxed{\Delta V_L = \mathcal{E}_0,\ \Delta V_C = 0,\ \Delta V_R = \mathcal{E}_0}$$

(b) After the switch has been closed a long time, the capacitor acts as an open switch. The steady-state conditions shown in ANS. FIG. CQ32.7 (b) will exist. The currents and voltages are:



ANS. FIG. CQ32.7(b)

$$\boxed{I_L = 0,\ I_C = 0,\ I_R = 0}$$

$$\boxed{\Delta V_L = 0,\ \Delta V_C = \mathcal{E}_0,\ \Delta V_R = 0}$$

CQ32.8 When the capacitor is fully discharged, the current in the circuit is a maximum. The inductance of the coil is making the current continue to flow. At this time the magnetic field of the coil contains all the

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energy that was originally stored in the charged capacitor. The current has just finished discharging the capacitor and is proceeding to charge it up again with the opposite polarity.

CQ32.9 According to Equations 32.31 and 32.32, the oscillator is overdamped if $R > R_C = \sqrt{\frac{4L}{C}}$: it will not oscillate. If $R < R_C$, then the oscillator is underdamped and can go through several cycles of oscillation before the current falls below background noise.

CQ32.10 The energy stored in a capacitor is proportional to the square of the electric field, and the energy stored in an induction coil is proportional to the square of the magnetic field. The capacitor's energy is proportional to its capacitance, which depends on its geometry and the dielectric material inside. The coil's energy is proportional to its inductance, which depends on its geometry and the core material. The capacitor's energy is proportional to the charge it stores, the coil's energy is proportional to the current it holds.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 32.1 Self-Induction and Induction

***P32.1** The magnitude of the average induced emf for this coil is

$$|\bar{\mathcal{E}}| = L\frac{\Delta i}{\Delta t} = \left(3.00\times10^{-3}\ \text{H}\right)\left(\frac{1.50\ \text{A} - 0.200\ \text{A}}{0.200\ \text{s}}\right) = 1.95\times10^{-2}\ \text{V}$$

$$= \boxed{19.5\ \text{mV}}$$

***P32.2** Treating the telephone cord as a solenoid, we have:

$$L = \frac{\mu_0 N^2 A}{\ell} = \frac{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)(70.0)^2\,\pi\left(6.50\times10^{-3}\ \text{m}\right)^2}{0.600\ \text{m}}$$

$$= \boxed{1.36\ \mu\text{H}}$$

P32.3 The self-induced emf at any instant is

$$\mathcal{E}_L = -L\frac{di}{dt}$$

Its average value is

$$\mathcal{E}_{L,\text{ave}} = -L\left(\frac{I_f - I_i}{t}\right) = (-2.00\ \text{H})\left(\frac{0 - 0.500\ \text{A}}{1.00\times10^{-2}\ \text{s}}\right)\left(\frac{\text{V}\cdot\text{s/A}}{1\ \text{H}}\right)$$

$$= \boxed{+100\ \text{V}}$$

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P32.4 (a) The inductance of the solenoid is

$$L = \frac{\mu_0 N^2 A}{\ell} = \frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)(400)^2\left[\pi\left(2.50\times 10^{-2}\ \text{m}\right)^2\right]}{0.200\ \text{m}}$$

$$= 1.97\times 10^{-3}\ \text{H} = \boxed{1.97\ \text{mH}}$$

(b) From $|\mathcal{E}| = L(\Delta i/\Delta t)$,

$$\frac{\Delta I}{\Delta t} = \frac{|\mathcal{E}|}{L} = \frac{75.0\times 10^{-6}\ \text{V}}{1.97\times 10^{-3}\ \text{H}} = 38.0\times 10^{-3}\ \text{A/s} = \boxed{38.0\ \text{mA/s}}$$

P32.5 From $\left|\mathcal{E}\right| = L\left(\frac{\Delta i}{\Delta t}\right)$, we have

$$L = \frac{\mathcal{E}}{(\Delta i/\Delta t)} = \frac{24.0\times 10^{-3}\ \text{V}}{10.0\ \text{A/s}} = 2.40\times 10^{-3}\ \text{H}$$

From $L = \frac{N\Phi_B}{i}$, we have

$$\Phi_B = \frac{Li}{N} = \frac{\left(2.40\times 10^{-3}\ \text{H}\right)(4.00\ \text{A})}{500}$$

$$= \boxed{19.2\ \mu\text{T}\cdot\text{m}^2}$$

P32.6 (a) $B = \mu_0 ni = \left(4\pi\times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(\frac{450}{0.120\ \text{m}}\right)(0.040\,0\ \text{A}) = \boxed{188\ \mu\text{T}}$

(b) $\Phi_B = BA = B\pi\left(\frac{15.0\times 10^{-3}\text{m}}{2}\right)^2 = \boxed{3.33\times 10^{-8}\ \text{T}\cdot\text{m}^2}$

(c) $L = \frac{N\Phi_B}{i} = \frac{450\Phi_B}{0.040\,0\ \text{A}} = \boxed{0.375\ \text{mH}}$

(d) $\boxed{B \text{ and } \Phi_B \text{ are proportional to current; } L \text{ is independent of current.}}$

P32.7 From $\left|\mathcal{E}\right| = L\left(\frac{\Delta i}{\Delta t}\right)$, we have

$$L = \frac{|\mathcal{E}|}{|\Delta i/\Delta t|} = \frac{|\mathcal{E}|(\Delta t)}{|\Delta i|} = \frac{\left(12.0\times 10^{-3}\ \text{V}\right)(0.500\ \text{s})}{|2.00\ \text{A} - 3.50\ \text{A}|}$$

$$= 4.00\times 10^{-3}\ \text{H} = \boxed{4.00\ \text{mH}}$$

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P32.8 (a) In terms of its cross-sectional area (A), length (ℓ), and number of turns (N), the self inductance of a solenoid is given as $L = \mu_0 N^2 A/\ell$. Thus, for the given solenoid,

$$L = \frac{\mu_0 N^2 \left(\pi d^2/4\right)}{\ell}$$

$$= \frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)(580)^2\left[\pi\left(8.00\times 10^{-2}\ \text{m}\right)^2/4\right]}{(0.36\ \text{m})}$$

$$= 5.90\times 10^{-3}\ \text{H} = \boxed{5.90\ \text{mH}}$$

(b) $$|\mathcal{E}| = \left|-L\left(\frac{\Delta i}{\Delta t}\right)\right| = \left(5.90\times 10^{-3}\ \text{H}\right)\left(+4.00\ \text{A/s}\right)$$

$$= 23.6\times 10^{-3}\ \text{V} = \boxed{23.6\ \text{mV}}$$

P32.9 $|\mathcal{E}| = L\frac{di}{dt} = \left|\left(90.0\times 10^{-3}\right)\frac{d}{dt}\left(1.00t^2 - 6.00t\right)\right| = (90.0)\left|(2.00t - 6.00)\right|$, where $\mathcal{E}$ is in millivolts (mV) and t in seconds.

(a) At t = 1.00 s, $\mathcal{E} = \boxed{360\ \text{mV}}$

(b) At t = 4.00 s, $\mathcal{E} = \boxed{180\ \text{mV}}$

(c) $\mathcal{E} = (90.0)(2t - 6) = 0$ when $\boxed{t = 3.00\ \text{s}}$

P32.10 The inductance is $L = \dfrac{\mu_0 N^2 A}{\ell}$ with $A = \pi r^2$. The induced emf as a function of time is $\mathcal{E}_L = -L\dfrac{di}{dt}$. By substitution we have

$$\mathcal{E}_L = -L\frac{di}{dt} = -\frac{\mu_0 N^2 \pi r^2}{\ell}\frac{di}{dt} \quad \text{and} \quad r = \left(\frac{-\mathcal{E}_L \ell}{\mu_0 N^2 \pi\, di/dt}\right)^{1/2}$$

Then $$r = \left(\frac{-(175\times 10^{-6}\ \text{V})(0.160\ \text{m})}{(4\pi\times 10^{-7}\ \text{N/A}^2)(420)^2\pi(-0.421\ \text{A/s})}\right)^{1/2} = \boxed{9.77\ \text{mm}}$$

P32.11 The emf is given by

$$\mathcal{E} = \mathcal{E}_0 e^{-kt} = -L\frac{di}{dt}$$

from which we obtain

$$di = -\frac{\mathcal{E}_0}{L}e^{-kt}dt$$

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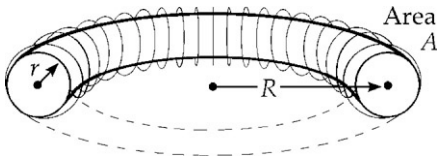
If we require $i \to 0$ as $t \to \infty$, the solution is $i = \dfrac{\mathcal{E}_0}{Lk} e^{-kt} = \dfrac{dq}{dt}$, so

$$Q = \int i\,dt = \int_0^\infty \frac{\mathcal{E}_0}{Lk} e^{-kt}\,dt = -\frac{\mathcal{E}_0}{Lk^2} \quad \rightarrow \quad \boxed{|Q| = \frac{\mathcal{E}_0}{Lk^2}}$$

P32.12 The inductance of a solenoid is $L = \dfrac{\mu_0 N^2 A}{\ell}$.

The long solenoid is bent into a circle of radius R, so its length $\ell \approx 2\pi R$; therefore, the inductance of the toroid is

$$L = \frac{\mu_0 N^2 A}{\ell} \approx \frac{\mu_0 N^2 (\pi r^2)}{2\pi R} = \frac{1}{2}\mu_0 N^2 \frac{r^2}{R}$$



ANS. FIG. P32.12

P32.13 Using the definition of self-inductance, $\mathcal{E} = -L\dfrac{di}{dt}$, we obtain

$$\mathcal{E} = -L\frac{d}{dt}(I_i \sin\omega t) = -L\omega(I_i \cos\omega t)$$
$$= -\left(10.0\times10^{-3}\right)\left[2\pi(60.0)\right](5.00)\cos\omega t$$

$$\boxed{\mathcal{E} = -18.8\cos 120\pi t, \text{ where } \mathcal{E} \text{ is in volts and } t \text{ is in seconds.}}$$

P32.14 The current change is linear, so $\mathcal{E} = -L\dfrac{di}{dt} = -L\dfrac{\Delta i}{\Delta t}$.

t = 0 to 4 ms:

$$\mathcal{E} = -(4.00\text{ mH})\frac{-2.00\text{ mA}}{4.00\text{ ms}} = +2.00\text{ mV}$$

t = 4 to 8 ms:

$$\mathcal{E} = -(4.00\text{ mH})\frac{+5.00\text{ mA}}{4.00\text{ ms}} = -5.00\text{ mV}$$

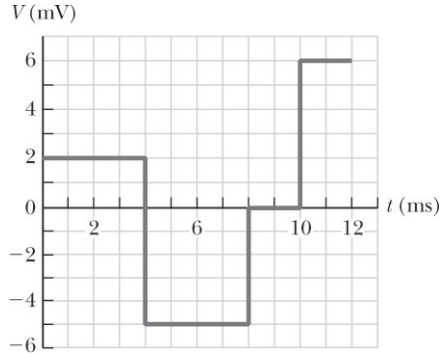
t = 8 to 10 ms:

$$\mathcal{E} = -(4.00\text{ mH})\frac{0}{2.00\text{ ms}} = 0.00\text{ mV}$$

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t = 10 to 12 ms:

$$\mathcal{E} = -(4.00\text{ mH})\frac{-3.00\text{ mA}}{2.00\text{ ms}} = +6.00\text{ mV}$$



ANS. FIG. P32.14

Section 32.2 *RL* Circuits

P32.15 (a) The inductance of a solenoid is

$$L = \frac{\mu_0 N^2 A}{\ell} = \frac{\mu_0 N^2 \pi r^2}{\ell} = \frac{\mu_0 (510)^2 \pi \left(8.00\times10^{-3}\text{ m}\right)^2}{0.140\text{ m}}$$
$$= 4.69\times10^{-4}\text{ H} = \boxed{0.469\text{ mH}}$$

(b) The time constant of the circuit is

$$\tau = \frac{L}{R} = \frac{4.69\times10^{-4}\text{ H}}{2.50\ \Omega} = 1.88\times10^{-4}\text{ s} = \boxed{0.188\text{ ms}}$$

P32.16 (a) At time t,

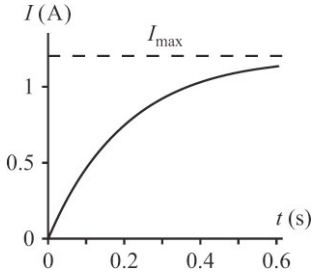
$$i(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right)$$

where

$$\tau = \frac{L}{R} = \frac{2.00\text{ H}}{10.0\ \Omega} = 0.200\text{ s}$$

After a long time,

$$I_i = \frac{\mathcal{E}}{R}\left(1 - e^{-\infty}\right) = \frac{\mathcal{E}}{R}$$



ANS. FIG. P32.16

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At $i(t) = 0.500I_i$

$$(0.500)\frac{\mathcal{E}}{R} = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right)$$

so $0.500 = 1 - e^{-t/\tau}$

Isolating the constants on the right,

$$e^{-t/\tau} = 0.500$$

$$\ln\left(e^{-t/\tau}\right) = \ln(0.500)$$

$$t = \tau\left[-\ln(0.500)\right] = (0.200\text{ s})\left[-\ln(0.500)\right] = \boxed{0.139\text{ s}}$$

(b) Similarly, to reach 90% of I_i, $0.900 = 1 - e^{-t/\tau} \quad \rightarrow \quad e^{-t/\tau} = 0.100$

and $t = -\tau\ln(0.100)$

Thus,

$$t = -(0.200\text{ s})\ln(0.100) = \boxed{0.461\text{ s}}$$

P32.17 (a) Using $\tau = RC = \frac{L}{R}$, we get

$$R = \sqrt{\frac{L}{C}} = \sqrt{\frac{3.00\text{ H}}{3.00\times10^{-6}\text{ F}}} = 1.00\times10^{3}\ \Omega = \boxed{1.00\text{ k}\Omega}.$$

(b) The time constant is

$$\tau = RC = \left(1.00\times10^{3}\ \Omega\right)\left(3.00\times10^{-6}\text{ F}\right)$$

$$= 3.00\times10^{-3}\text{ s} = \boxed{3.00\text{ ms}}$$

P32.18 The current builds exponentially according to:

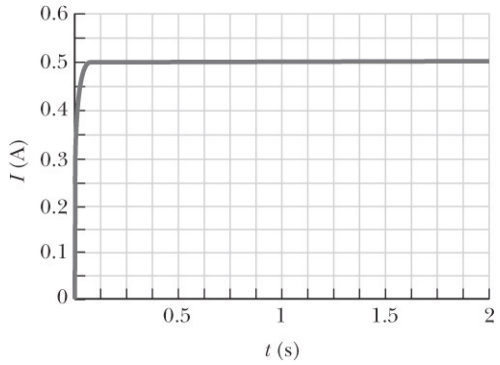
$$i(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right) = \frac{12.0\text{ V}}{24.0\ \Omega}\left(1 - e^{-t/\tau}\right)$$

$$= 0.500\left(1 - e^{-t/\tau}\right)$$

where current I is in amperes (A) and time t is in seconds (s).
The current increases from 0 to asymptotically approach 0.500 A. In case (a) the current jumps up essentially instantaneously. In case (b) it increases with a longer time constant, and in case (c) the increase is still slower.

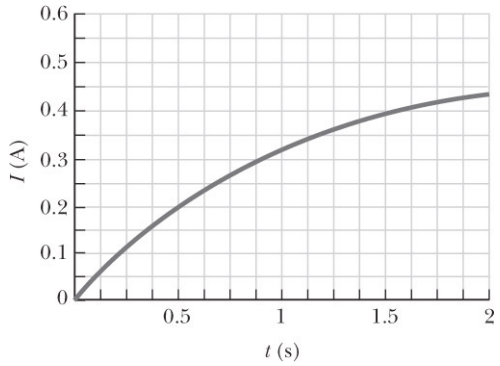
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(a) With "essentially zero" inductance, we take $\tau = \frac{L}{R} = 0.01$. ANS. FIG. P32.18(a) graphs $I(t)$ for this case.



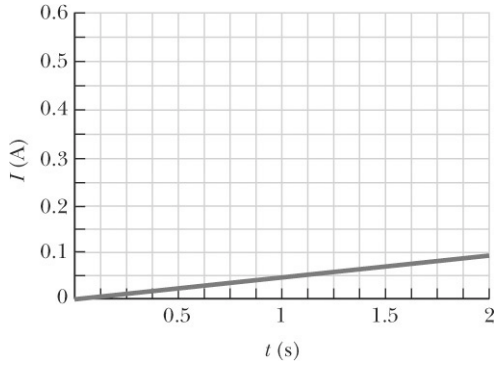
ANS. FIG. P32.18(a)

(b) We take $\tau = \frac{L}{R} = 1$. ANS. FIG. P32.18(b) graphs $I(t)$ for this case.



ANS. FIG. P32.18(b)

(c) We take $\tau = \frac{L}{R} = 10$. ANS. FIG. P32.18(c) graphs $I(t)$ for this case.



ANS. FIG. P32.18(c)

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P32.19 (a) The two resistors are in parallel. Their resistance values 450 Ω and R are related to their equivalent resistance R_{eq} by

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{450\,\Omega}$$

and the equivalent resistance is related to the time constant of the circuit by

$$\tau = \frac{L}{R_{eq}} \rightarrow \frac{1}{R_{eq}} = \frac{\tau}{L}$$

Thus,

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{450\ \Omega} = \frac{\tau}{L}$$

Solving for R,

$$\frac{1}{R} = \frac{\tau}{L} - \frac{1}{450\ \Omega} = \frac{15.0\times10^{-6}\ \text{s}}{5.00\times10^{-3}\ \text{H}} - \frac{1}{450\ \Omega}$$

which gives

$$R = 1\ 290\ \Omega = \boxed{1.29\ \text{k}\Omega}$$

(b) The current will immediately begin to die from the value it had just before the switch was thrown to position b. Before the switch position was changed, the current was constant in time, so there was no emf induced in the inductor. The current was just

$$i = \frac{\Delta V}{R_{eq}} = \Delta V\frac{\tau}{L} = (24.0\ \text{V})\frac{15.0\times10^{-6}\ \text{s}}{5.00\times10^{-3}\ \text{H}}$$

$$= 0.072\ 0\ \text{A} = \boxed{72.0\ \text{mA}}$$

***P32.20** The current increases as a function of time as

$$i = I_i\left(1 - e^{-t/\tau}\right)$$

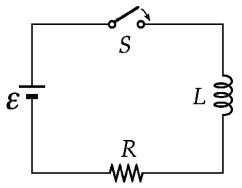
Substituting,

$$0.980 = 1 - e^{-3.00\times10^{-3}/\tau}$$

$$0.020\,0 = e^{-3.00\times10^{-3}/\tau}$$

Solving for the time constant gives

$$\tau = -\frac{3.00\times10^{-3}}{\ln(0.020\,0)} = 7.67\times10^{-4}\ \text{s}$$



ANS. FIG. P32.20

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and since $\tau = \frac{L}{R}$,

$$L = \tau R = (7.67 \times 10^{-4}\ \text{s})(10.0\ \Omega) = \boxed{7.67\ \text{mH}}$$

***P32.21** For the increasing current $i = \frac{\mathcal{E}}{R}\left(1 - e^{-Lt/R}\right)$. The final value is $\frac{\mathcal{E}}{R}$, so the condition on Δt is

$$0.800\frac{\mathcal{E}}{R} = \frac{\mathcal{E}}{R}\left(1 - e^{-L\Delta t/R}\right)$$

$$e^{-L\Delta t/R} = 0.200$$

$$e^{+L\Delta t/R} = 5.00$$

$$\frac{L\Delta t}{R} = \ln 5.00$$

$$\Delta t = \frac{R \ln 5.00}{L}$$

At the moment when the battery is removed, the current in the coil is quite precisely $\frac{\mathcal{E}}{R}$. During the decrease, $i = I_i e^{-Lt/R} = \frac{\mathcal{E}}{R} e^{-Lt/R}$.

(a) at $t = \Delta t = \frac{R \ln 5.00}{L}$,

$$\frac{i}{I_i} = e^{-L\Delta t/R} = 0.200 = \boxed{20.0\%}$$

(b) at $t = 2\Delta t$,

$$\frac{i}{I_i} = e^{-L2\Delta t/R} = \left(e^{-L\Delta t/R}\right)^2 = (0.200)^2 = 0.040\ 0 = \boxed{4.00\%}$$

P32.22 Taking $\tau = \frac{L}{R}$, and $i = I_i e^{-t/\tau}$: $\frac{di}{dt} = I_i e^{-t/\tau}\left(-\frac{1}{\tau}\right)$

$iR + L\frac{di}{dt} = 0$ will be true if $I_i R\ e^{-t/\tau} + L\left(I_i e^{-t/\tau}\right)\left(-\frac{1}{\tau}\right) = 0$

We have agreement because $\tau = \frac{L}{R}$.

P32.23 The current at this time is given by

$$i = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right) = \frac{120\ \text{V}}{9.00\ \Omega}\left[1 - e^{-0.200/(7.00/9.00)}\right] = 3.02\ \text{A}$$

Then, $\Delta V_R = iR = (3.02)(9.00) = 27.2\ \text{V}$

and $\Delta V_L = \mathcal{E} - \Delta V_R = 120 - 27.2 = \boxed{92.8\ \text{V}}$

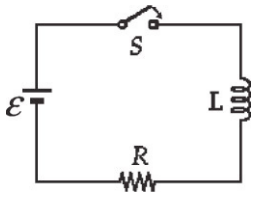
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P32.24 (a) $\tau = \dfrac{L}{R} = \dfrac{8.00\times10^{-3}\ \text{H}}{4.00\ \Omega} = 2.00\times10^{-3}\ \text{s} = \boxed{2.00\ \text{ms}}$

(b) $i = \dfrac{\mathcal{E}}{R}\left(1-e^{-t/\tau}\right) = \left(\dfrac{6.00\ \text{V}}{4.00\ \Omega}\right)\left(1-e^{-0.250/2.00}\right) = \boxed{0.176\ \text{A}}$

(c) $I_i = \dfrac{\mathcal{E}}{R} = \dfrac{6.00\ \text{V}}{4.00\ \Omega} = \boxed{1.50\ \text{A}}$

(d) $0.800 = 1-e^{-t/2.00\ \text{ms}} \rightarrow t = -(2.00\ \text{ms})\ln(0.200) = \boxed{3.22\ \text{ms}}$



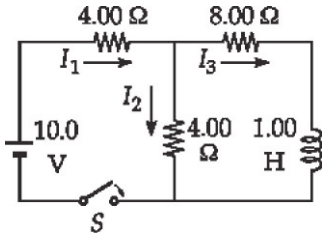
ANS. FIG. P32.24

P32.25 Name the currents as shown in ANS. FIG. P32.25. By Kirchhoff's laws:

$$i_1 = i_2 + i_3 \qquad \textbf{[1]}$$

$$+10.0\ \text{V} - 4.00i_1 - 4.00i_2 = 0 \qquad \textbf{[2]}$$

$$+10.0\ \text{V} - 4.00i_1 - 8.00i_3 - (1.00)\frac{di_3}{dt} = 0 \qquad \textbf{[3]}$$



ANS. FIG. P32.25

From [1] and [2],

$$+10.0 - 4.00i_1 - 4.00i_1 + 4.00i_3 = 0$$

$$i_1 = 0.500i_3 + 1.25\ \text{A}$$

Then [3] becomes $10.0\ \text{V} - 4.00(0.500i_3 + 1.25\ \text{A}) - 8.00i_3 - (1.00)\dfrac{di_3}{dt} = 0$

$$(1.00\ \text{H})\left(\frac{di_3}{dt}\right) + (10.0\ \Omega)i_3 = 5.00\ \text{V}$$

or $$5.00\ \text{V} - (10.0\ \Omega)i_3 - (1.00\ \text{H})\left(\frac{di_3}{dt}\right) = 0$$

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which can be compared to the general form (Equation 32.6)

$$\mathcal{E} - iR - L\frac{di}{dt} = 0$$

which has the solution (from Equation 32.7) $i = \frac{\mathcal{E}}{R}\left(1 - e^{-Rt/L}\right)$.

Thus, we have:

(a) $i_3 = \left(\frac{5.00\text{ V}}{10.0\ \Omega}\right)\left[1 - e^{-(10.0\ \Omega)t/1.00\text{ H}}\right] = \boxed{(0.500\text{ A})\left[1 - e^{-10t/\text{s}}\right]}$

(b) $i_1 = 1.25 + 0.500I_3 = \boxed{1.50\text{ A} - (0.250\text{ A})e^{-10t/\text{s}}}$

P32.26 Refer to ANS. FIG. P32.25 above. Name the currents as shown. By Kirchhoff's laws:

$$i_1 = i_2 + i_3 \quad \textbf{[1]}$$

$$\mathcal{E} - Ri_1 - Ri_2 = 0 \quad \textbf{[2]}$$

$$\mathcal{E} - Ri_1 - 2Ri_3 - L\frac{di_3}{dt} = 0 \quad \textbf{[3]}$$

From [1] and [2],

$$\mathcal{E} - Ri_1 - R\left(i_1 - i_3\right) = 0$$
$$\mathcal{E} - Ri_1 - Ri_1 + Ri_3 = 0$$
$$i_1 = \frac{1}{2}i_3 + \frac{\mathcal{E}}{2R}$$

Then [3] becomes

$$\mathcal{E} - R\left(\frac{1}{2}i_3 + \frac{\mathcal{E}}{2R}\right) - 2Ri_3 - L\frac{di_3}{dt} = 0$$
$$L\frac{di_3}{dt} + 2.5Ri_3 = \frac{\mathcal{E}}{2}$$
$$\frac{\mathcal{E}}{2} - 2.5Ri_3 - L\frac{di_3}{dt} = 0$$

which can be compared to the general form (Equation 32.6)

$$\mathcal{E} - iR - L\frac{di}{dt} = 0$$

which has the solution (from Equation 32.7)

$$i = \frac{\mathcal{E}}{R}\left(1 - e^{-Rt/L}\right)$$

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Thus, we have:

(a) $$i_3 = \left(\frac{\mathcal{E}/2}{2.5R}\right)\left[1 - e^{-2.5Rt/L}\right] = \boxed{\frac{\mathcal{E}}{5R}\left(1 - e^{-5Rt/2L}\right)}$$

(b) $$i_1 = \frac{1}{2}i_3 + \frac{\mathcal{E}}{2R} = \frac{1}{2}\left[\frac{\mathcal{E}}{5R}\left(1 - e^{-5Rt/2L}\right)\right] + \frac{\mathcal{E}}{2R}$$

$$i_1 = \frac{\mathcal{E}}{10R}\left(1 - e^{-5Rt/2L}\right) + \frac{5\mathcal{E}}{10R} = \boxed{\frac{\mathcal{E}}{10R}\left(6 - e^{-5Rt/2L}\right)}$$

P32.27 (a) When $i = 2.00$ A, the voltage across the resistor is

$$\Delta V_R = iR = (2.00\ \text{A})(8.00\ \Omega) = 16.0\ \text{V}$$

Kirchhoff's loop rule tells us that the sum of the changes in potential around the loop must be zero:

$$\mathcal{E} - \Delta V_R - \mathcal{E}_L = 36.0\ \text{V} - 16.0\ \text{V} - \mathcal{E}_L = 0$$

so $\mathcal{E}_L = 20.0$ V

and $$\frac{\Delta V_R}{\mathcal{E}_L} = \frac{16.0\ \text{V}}{20.0\ \text{V}} = \boxed{0.800}$$

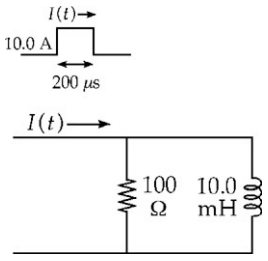
(b) Similarly, for $i = 4.50$ A, $\Delta V_R = iR = (4.50\ \text{A})(8.00\ \Omega) = 36.0$ V and

$$\mathcal{E} - \Delta V_R - \mathcal{E}_L = 36.0\ \text{V} - 36.0\ \text{V} - \mathcal{E}_L = 0$$

so $\mathcal{E}_L = \boxed{0}$

P32.28 For $t \le 0$, the current in the inductor is $\boxed{\text{zero}}$.

For $0 \le t \le 200\ \mu$s, there will be current i_R in the resistor and I_L in the inductor so that $i = i_R + i_L = I_i = 10.0$ A. Assuming both currents are downward in ANS. FIG. P32.28, we apply Kirchhoff's loop rule going counterclockwise around the loop, and we find that



ANS. FIG. P32.28

$$-i_R R + L\frac{di_L}{dt} = 0$$

Using $I_i = i_R + i_L \rightarrow i_R = I_i - i_L$, we have

$$-(I_i - i_L)R + L\frac{di_L}{dt} = 0$$

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Then,

$$L\frac{dI_L}{dt} = \left(I_{\max} - I_L\right)R$$

$$\int_0^I \frac{di_L}{\left(I_i - I_L\right)} = \int_0^t \frac{R}{L}dt$$

$$-\ln\frac{\left(I_\text{i} - i_L\right)}{I_i} = \frac{R}{L}t$$

which gives

$$i_L = I_i\left(1 - e^{-Rt/L}\right)$$

We see that $t = 0$, $i_L = 0$ as we expect because of the back emf induced in the inductor. With the time constant

$$\tau = \frac{L}{R} = \frac{(10.0\text{ mH})}{(100\ \Omega)} = 1.00\times10^{-4}\text{ s}$$

we have

$$i_L = I_i\left(1 - e^{-t/\tau}\right) = \boxed{(10.0\text{ A})\left(1 - e^{-10\,000t/\text{s}}\right)} \qquad (0 \le t \le 200\ \mu\text{s})$$

At $t = 200\ \mu$s, $i = (10.00\text{ A})\left(1 - e^{-2.00}\right) = 8.65\text{ A}$; thereafter, the current decays. The loop rule gives the same result,

$$-i_R R + L\frac{di_L}{dt} = 0$$

but now $i_R + i_L = 0 \rightarrow i_R = -i_L$, so we have

$$i_L R + L\frac{di_L}{dt} = 0 \rightarrow L\frac{di_L}{dt} = -i_L R$$

$$\int_{I_i}^I \frac{di_L}{i_L} = -\int_{200\ \mu\text{s}}^t \frac{R}{L}dt$$

$$\ln\frac{i_L}{I_i} = -\frac{R}{L}(t - 200\ \mu\text{s}) \rightarrow i_L = I_i e^{-R(t-200\,\mu\text{s})/L}$$

For $t = 200\ \mu$s, $i_i = 8.65$ A, and for $t \ge 200\ \mu$s,

$$i = (8.65\text{ A})e^{-10\,000(t-200\ \mu\text{s})/\text{s}} = (8.65\text{ A})e^{-10\,000t/\text{s}+2.00}$$

$$= \left(8.65e^{2.00}\text{ A}\right)e^{-10\,000t/\text{s}} = \boxed{(63.9\text{ A})e^{-10\,000t/\text{s}}} \qquad (t \ge 200\ \mu\text{s})$$

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P32.29 From Equation 32.7, $i = I_i\left(1 - e^{-t/\tau}\right)$. Therefore,

$$\frac{di}{dt} = -I_i\left(e^{-t/\tau}\right)\left(-\frac{1}{\tau}\right)$$

where

$$\tau = \frac{L}{R} = \frac{15.0\ \text{H}}{30.0\ \Omega} = 0.500\ \text{s}$$

Then, $\frac{di}{dt} = \frac{R}{L} I_i e^{-t/\tau}$ with $I_i = \frac{\mathcal{E}}{R}$

(a) At $t = 0$,

$$\frac{di}{dt} = \frac{R}{L} I_i e^0 = \frac{\mathcal{E}}{L} = \frac{100\ \text{V}}{15.0\ \text{H}} = \boxed{6.67\ \text{A/s}}$$

(b) At $t = 1.50$ s,

$$\frac{di}{dt} = \frac{\mathcal{E}}{L} e^{-t/\tau} = \left(6.67\ \text{A/s}\right) e^{-1.50/(0.500)} = \left(6.67\ \text{A/s}\right) e^{-3.00}$$

$$= \boxed{0.332\ \text{A/s}}$$

P32.30 (a) For a series connection, both inductors carry equal currents at every instant, so $\frac{di}{dt}$ is the same for both. The voltage across the pair is

$$L_{\text{eq}} \frac{di}{dt} = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} \quad \rightarrow \quad L_{\text{eq}} = L_1 + L_2$$

(b) For a parallel connection, the voltage across each inductor is the same for both.

$$L_{\text{eq}} \frac{di}{dt} = L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt} = \Delta V_L$$

where the currents are related by $i = i_1 + i_2$. Therefore,

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$$

$$\frac{\Delta V_L}{L_{\text{eq}}} = \frac{\Delta V_L}{L_1} + \frac{\Delta V_L}{L_2} \quad \rightarrow \quad \frac{1}{L_{\text{eq}}} = \frac{1}{L_1} + \frac{1}{L_2}$$

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(c) $L_{eq}\frac{di}{dt} + R_{eq}i = L_1\frac{di}{dt} + iR_1 + L_2\frac{di}{dt} + iR_2$

Now i and $\frac{di}{dt}$ are independent quantities under our control, so functional equality requires both $L_{eq} = L_1 + L_2$ and $R_{eq} = R_1 + R_2$.

(d) Yes. The relations $\Delta V = L_{eq}\frac{di}{dt} + R_{eq}i = L_1\frac{di_1}{dt} + R_1 i_1 = L_2\frac{di_2}{dt} + R_2 i_2$, where $i = i_1 + i_2$ and $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$, must always be true.

We may choose to keep the currents constant in time. Then, from $i = i_1 + i_2$, we have

$$\frac{\Delta V_L}{R_{eq}} = \frac{\Delta V_L}{R_1} + \frac{\Delta V_L}{R_2} \quad \rightarrow \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

We may choose to make the current oscillate so that at a given moment it is zero. Then, from $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$, as in part (b), we have $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$.

P32.31 (a) The equation for current buildup is obtained by combining Equations 32.7 and 32.8:

$$i = \frac{\mathcal{E}\left(1 - e^{-Rt/L}\right)}{R}$$

We proceed step-by-step to solve for t in terms of the other quantities, all of which are given:

$$iR/\mathcal{E} = 1 - e^{-Rt/L}$$

so $e^{-Rt/L} = 1 - iR/\mathcal{E}$

and $-Rt/L = \ln(1 - iR/\mathcal{E})$

then,

$$t = -(L/R)\ln(1 - iR/\mathcal{E})$$

$$t = -(0.140\ \text{H}/4.90\ \Omega)\ln[1 - (0.220\ \text{A})(4.90\ \Omega)/6.00\ \text{V}]$$

$$= -(0.028\ 6\ \text{s})\ln(0.820)$$

$$t = -(0.028\ 6\ \text{s})(-0.198) = \boxed{5.66\ \text{ms}}$$

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(b) We now make the general equation refer to a different instant. The current after ten seconds is

$$i = \left(\frac{6.00\ \text{V}}{4.90\ \Omega}\right)\left(1 - e^{(-35.0\ \text{s}^{-1})(10.0\ \text{s})}\right) = (1.22\ \text{A})(1 - e^{-350}) = \boxed{1.22\ \text{A}}$$

(c) The equation for current decrease after the battery is removed is $i = \frac{\mathcal{E}}{R}e^{-Rt/L}$. We solve for t:

$$\frac{iR}{\mathcal{E}} = e^{-Rt/L} \quad \text{or} \quad \frac{\mathcal{E}}{iR} = e^{+Rt/L}$$

Then,

$$\ln\left(\mathcal{E}/iR\right) = Rt/L \quad \text{and} \quad t = \left(L/R\right)\ln\left(\mathcal{E}/iR\right)$$

Substituting,

$$t = (0.140\ \text{H}/4.90\ \Omega)\ \ln[6.00\ \text{V}/(0.160\ \text{A} \cdot 4.90\ \Omega)]$$
$$= (0.028\ 6\ \text{s})(\ln 7.65) = \boxed{58.1\ \text{ms}}$$

Section 32.3 Energy in a Magnetic Field

P32.32 The inductance of the solenoid is

$$L = N\frac{\Phi_B}{i} = 200\frac{3.70 \times 10^{-4}\ \text{Wb}}{1.75\ \text{A}} = 0.042\ 3\ \text{H}$$

The energy stored is

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}(0.042\ 3\ \text{H})(1.75\ \text{A})^2 = 0.064\ 8\ \text{J} = \boxed{64.8\ \text{mJ}}$$

P32.33 For a solenoid of length ℓ, the inductance is $L = \frac{\mu_0 N^2 A}{\ell}$.

Thus, since $U_B = \frac{1}{2}Li^2 = \frac{\mu_0 N^2 A i^2}{2\ell}$, the stored energy is

$$U_B = \frac{(4\pi \times 10^{-7}\ \text{N/A}^2)(68)^2\pi(6.00 \times 10^{-3}\ \text{m})^2(0.770\ \text{A})^2}{2\ (0.080\ 0)\ \text{m}}$$
$$= \boxed{2.44 \times 10^{-6}\ \text{J}}$$

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P32.34 From Equation 32.7, $i = \frac{\mathcal{E}}{R}\left(1 - e^{-Rt/L}\right)$:

(a) The maximum current, after a long time t, is $i = I_i = \frac{\mathcal{E}}{R} = 2.00\ \text{A}$.

At that time, the inductor is fully energized and

$$P = i(\Delta V) = (2.00\ \text{A})(10.0\ \text{V}) = \boxed{20.0\ \text{W}}$$

(b) $P_{\text{lost}} = i^2 R = (2.00\ \text{A})^2(5.00\ \Omega) = \boxed{20.0\ \text{W}}$

(c) The inductor has no resistance: $P_{\text{inductor}} = i\left(\Delta V_{\text{drop}}\right) = \boxed{0}$

(d) $U_B = \frac{1}{2}Li^2 = \frac{1}{2}(10.0\ \text{H})(2.00\ \text{A})^2 = \boxed{20.0\ \text{J}}$

P32.35 (a) The energy density stored by the electric field is

$$u_E = \epsilon_0 \frac{E^2}{2} = \left(8.85 \times 10^{-12}\ \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}\right)\frac{(100\ \text{V/m})^2}{2}\left(\frac{\text{J/C}}{\text{V}}\right)^2\left(\frac{\text{N}\cdot\text{m}}{\text{J}}\right)$$

$$= 4.43 \times 10^{-8}\ \frac{\text{J}}{\text{m}^3} = \boxed{44.3\ \text{nJ/m}^3}$$

(b) The energy density stored by the magnetic field is

$$u_B = \frac{B^2}{2\mu_0} = \frac{\left(0.500 \times 10^{-4}\ \text{T}\right)^2}{2\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)}\left(\frac{\text{N/A}\cdot\text{m}}{\text{T}}\right)$$

$$= 9.95 \times 10^{-4}\ \frac{\text{N}}{\text{m}^2}\left(\frac{\text{m}}{\text{m}}\right) = 9.95 \times 10^{-4}\ \frac{\text{N}\cdot\text{m}}{\text{m}^3} = \boxed{995\ \mu\text{J/m}^3}$$

P32.36 We compute the integral:

$$\int_0^\infty e^{-2Rt/L}\,dt = -\frac{L}{2R}\int_0^\infty e^{-2Rt/L}\left(\frac{-2R\,dt}{L}\right) = -\frac{L}{2R}e^{-2Rt/L}\Big|_0^\infty$$

$$= -\frac{L}{2R}\left(e^{-\infty} - e^0\right) = \frac{L}{2R}(0 - 1) = \boxed{\frac{L}{2R}}$$

***P32.37** The current in the circuit at time t is $i = I_i\left(1 - e^{-t/\tau}\right)$, where $I_i = \frac{\mathcal{E}}{R}$, and the energy stored in the inductor is $U_B = \frac{1}{2}Li^2$.

(a) As $t \to \infty$, $I \to I_i = \frac{\mathcal{E}}{R} = \frac{24.0\ \text{V}}{8.00\ \Omega} = 3.00\ \text{A}$, and

$$U_B = \frac{1}{2}LI_i^2 = \frac{1}{2}(4.00\ \text{H})(3.00\ \text{A})^2 = \boxed{18.0\ \text{J}}$$

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(b) At $t = \tau$, $I = I_i\left(1 - e^{-1}\right) = (3.00\text{ A})(1 - 0.368) = 1.90\text{ A}$, and

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}(4.00\text{ H})(1.90\text{ A})^2 = \boxed{7.19\text{ J}}$$

P32.38 (a) $P = i\Delta V = (3.00\text{ A})(22.0\text{ V}) = \boxed{66.0\text{ W}}$

(b) $P = i\Delta V_R = i^2R = (3.00\text{ A})^2(5.00\ \Omega) = \boxed{45.0\text{ W}}$

(c) METHOD 1: We treat the real inductor as an ideal inductor (with no resistance) in series with an ideal resistor (with no inductance). When the current is 3.00 A, Kirchhoff's loop rule reads

$$+22.0\text{ V} - (3.00\text{ A})(5.00\ \Omega) - \Delta V_L = 0$$
$$\Delta V_L = 7.00\text{ V}$$

The power being stored in the inductor is

$$i\Delta V_L = (3.00\text{ A})(7.00\text{ V}) = \boxed{21.0\text{ W}}$$

METHOD 2: We do not treat the real inductor as an ideal inductor in series with an ideal resistor.

We wish to find the rate at which energy is being delivered to the inductor. As discussed in Section 32.3, $U_B = \frac{1}{2}Li^2 \rightarrow \frac{dU_B}{dt} = Li\frac{di}{dt}$. We know L (0.040 0 H) and i (3.00 A); we need to evaluate the term $\frac{di}{dt}$. From Equations 32.7 and 32.8 (or Equation 32.9),

$$i = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right) \rightarrow \frac{di}{dt} = \frac{\mathcal{E}}{L}e^{-t/\tau}$$

because $\tau = \frac{L}{R}$. Also,

$$i = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau}\right) \rightarrow e^{-t/\tau} = 1 - \frac{iR}{\mathcal{E}}$$

Therefore,

$$\frac{dU_B}{dt} = Li\frac{di}{dt} = Li\left(\frac{\mathcal{E}}{L}e^{-t/\tau}\right) = i\mathcal{E}e^{-t/\tau} = i\mathcal{E}\left(1 - \frac{iR}{\mathcal{E}}\right) = i(\mathcal{E} - iR)$$

When i = 3.00 A,

$$\frac{dU_B}{dt} = i(\mathcal{E} - iR) = (3.00\text{ A})\left[22.0\text{ V} - (3.00\text{ A})(5.00\ \Omega)\right]$$
$$= \boxed{21.0\text{ W}}$$

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(d) The power supplied by the battery is equal to the sum of the power delivered to the internal resistance of the coil and the power stored in the magnetic field.

(e) Yes.

(f) Just after $t = 0$, the current is very small, so the power delivered to the internal resistance of the coil (iR^2) is nearly zero, but the rate of the change of the current is large, so the power delivered to the magnetic field (Ldi/dt) is large, and nearly all the battery power is being stored in the magnetic field. Long after the connection is made, the current is not changing, so no power is being stored in the magnetic field, and all the battery power is being delivered to the internal resistance of the coil.

P32.39 (a) The magnetic energy density is given by

$$u_B = \frac{B^2}{2\mu_0} = \frac{(4.50\ \text{T})^2}{2\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)} = 8.06 \times 10^6\ \text{J/m}^3 = \boxed{8.06\ \text{MJ}}$$

(b) The magnetic energy stored in the field equals u times the volume of the solenoid (the volume in which B is non-zero).

$$U_B = u_B V = \left(8.06 \times 10^6\ \text{J/m}^3\right)\left[(0.260\ \text{m})\pi\left(0.031\,0\ \text{m}\right)^2\right]$$

$$= \boxed{6.32\ \text{kJ}}$$

Section 32.4 Mutual Inductance

P32.40 We use Equation 32.17, $|\mathcal{E}_2| = \left|-M\frac{di_1}{dt}\right|$, from which we obtain the mutual inductance:

$$M = \frac{|\mathcal{E}_2|}{|di_1/dt|} = \frac{0.096\,0\ \text{V}}{1.20\ \text{A/s}} = 0.080\ 0\ \text{H} = \boxed{80.0\ \text{mH}}$$

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P32.41 Let the changing current in coil 1 induce an emf in coil 2. Then,

$$\begin{aligned}\mathcal{E}_2 &= -M\frac{di_1}{dt} = -\left(100\times10^{-6}\right)\frac{d}{dt}\left[10.0\sin\left(1.00\times10^3 t\right)\right]\\ &= -\left(100\times10^{-6}\right)(10.0)\left(1.00\times10^3\right)\cos\left(1.00\times10^3 t\right)\\ &= -(1.00)\cos\left(1.00\times10^3 t\right)\end{aligned}$$

Therefore, the peak emf is $\left(\mathcal{E}_2\right)_{\max} = \boxed{1.00\text{ V}}$.

P32.42 The current is given by $i = I_i e^{-\alpha t}\sin\omega t,$ with $I_i = 5.00,\ \alpha = 0.025\,0$, and $\omega = 120\pi$. Then,

$$\begin{aligned}\frac{di}{dt} &= \frac{d}{dt}\left[I_i e^{-\alpha t}\sin\omega t\right]\\ &= I_i\left(-\alpha e^{-\alpha t}\right)\sin\omega t + I_i e^{-\alpha t}\left(\omega\cos\omega t\right)\\ &= I_i e^{-\alpha t}\left(-\alpha\sin\omega t + \omega\cos\omega t\right)\end{aligned}$$

where $\frac{di}{dt}$ is in amperes per second, I_i is in amperes, and t in seconds.

At $t = 0.800$ s,

$$\begin{aligned}\frac{di}{dt} &= (5.00)e^{-0.020\,0}\left\{-\left(0.025\,0\right)\sin\left[0.800(120\pi)\right]\right.\\ &\qquad\left.+\,120\pi\cos\left[0.800(120\pi)\right]\right\}\\ &= 1.85\times10^3\ \text{A/s}\end{aligned}$$

Thus, from $\mathcal{E}_2 = -M\frac{di_1}{dt}$,

$$M = \frac{-\mathcal{E}_2}{di_1/dt} = \frac{+3.20\text{ V}}{1.85\times10^3\ \text{A/s}} = \boxed{1.73\text{ mH}}$$

P32.43 (a) The mutual inductance of the coils is

$$M = \frac{N_B\Phi_{BA}}{i_A} = \frac{700\left(90.0\times10^{-6}\,\text{Wb}\right)}{3.50\text{ A}} = \boxed{18.0\text{ mH}}$$

(b) The inductance of coil A is

$$L_A = \frac{\Phi_A}{i_A} = \frac{400\left(300\times10^{-6}\,\text{Wb}\right)}{3.50\text{ A}} = \boxed{34.3\text{ mH}}$$

(c) The emf induced in the other coil is

$$\left|\mathcal{E}_B\right| = \left|-M\frac{di_A}{dt}\right| = (18.0\text{ mH})\left(0.500\ \text{A/s}\right) = \boxed{9.00\text{ mV}}$$

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P32.44 (a) Solenoid S_1 creates a nearly uniform field everywhere inside it, given by $B_1 = \mu_0 N_1 i/\ell$. The flux through one turn of solenoid S_2 is

$$\mu_0 \pi R_2^2 N_1 i/\ell$$

The emf induced in solenoid S_2 is

$$-(\mu_0 \pi R_2^2 N_1 N_2/\ell)(di/dt)$$

The mutual inductance is

$$\boxed{M_{12} = \mu_0 \pi R_2^2 N_1 N_2/\ell}$$

(b) Solenoid S_2 creates a nearly uniform field everywhere inside it, given by $B_2 = \mu_0 N_2 i_2/\ell$ and nearly zero field outside. The flux through one turn of solenoid 1 is

$$\mu_0 \pi R_2^2 N_2 i_2/\ell$$

The emf induced in solenoid 1 is

$$-(\mu_0 \pi R_2^2 N_1 N_2/\ell)(di_2/dt)$$

The mutual inductance is

$$\boxed{M_{12} = \mu_0 \pi R_2{}^2 N_1 N_2/\ell.}$$

(c) $\boxed{\text{They are the same.}}$

P32.45 Assume the long wire carries current I. Then the magnitude of the magnetic field it generates at distance x from the wire is $B = \dfrac{\mu_0 I}{2\pi x}$, and this field passes perpendicularly through the plane of the loop. The flux through the loop is

$$\Phi_B = \int \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = \int B\, dA = \int B(\ell\, dx) = \frac{\mu_0 I \ell}{2\pi} \int_{0.400\text{ mm}}^{1.70\text{ mm}} \frac{dx}{x}$$

$$= \frac{\mu_0 I \ell}{2\pi} \ln\left(\frac{1.70}{0.400}\right)$$

The mutual inductance between the wire and the loop is then

$$M = \frac{N_2 \Phi_{12}}{I_1} = \frac{N_2 \mu_0 \cancel{I} \ell}{2\pi \cancel{I}} \ln\left(\frac{1.70}{0.400}\right)$$

$$= \frac{1\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(2.70 \times 10^{-3}\ \text{m}\right)}{2\pi} \ln\left(\frac{1.70}{0.400}\right)$$

$$M = 7.81 \times 10^{-10}\ \text{H} = \boxed{781\ \text{pH}}$$

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P32.46 (a) A current i in the large loop of radius R produces a magnetic field of magnitude $B = \frac{\mu_0 i}{2R}$ at its center. Because the radius of the small loop $r << R$, we may treat the flux through the small loop as being approximately $\Phi_B = BA\cos 0.00° = \left(\frac{\mu_0 i}{2R}\right)A = \frac{\mu_0 \pi r^2 i}{2R}$. The mutual inductance of the loops is then

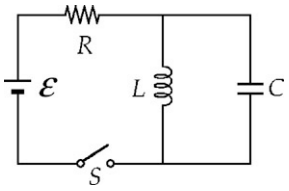
$$M = \frac{\Phi_B}{i} = \frac{\mu_0 \pi r^2}{2R}$$

(b) $$M = \frac{\mu_0 \pi r^2}{2R} = \frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\pi(0.020\ 0\ \text{m})^2}{2(0.200\ \text{m})} = 3.95 \times 10^{-9}\ \text{H}$$

$$= \boxed{3.95\ \text{nH}}$$

Section 32.5 Oscillations in an *LC* Circuit

P32.47 When the switch has been closed for a long time, battery, resistor, and coil carry constant current $I_i = \frac{\mathcal{E}}{R}$. When the switch is opened, current in battery and resistor drops to zero, but the coil carries this same current for a moment as oscillations begin in the *LC* loop.



ANS. FIG. P32.47

We interpret the problem to mean that the voltage amplitude of these oscillations is ΔV, in $\frac{1}{2}C(\Delta V)^2 = \frac{1}{2}LI_i^2$.

Then,

$$L = \frac{C(\Delta V)^2}{I_i^2} = \frac{C(\Delta V)^2 R^2}{\mathcal{E}^2} = \frac{\left(0.500 \times 10^{-6}\ \text{F}\right)(150\ \text{V})^2(250\ \Omega)^2}{(50.0\ \text{V})^2}$$

$$= 0.281\ \text{H} = \boxed{281\ \text{mH}}$$

P32.48 This radio is a radiotelephone on a ship, according to frequency assignments made by international treaties, laws, and decisions of the National Telecommunications and Information Administration.

The resonance frequency is $f_0 = \frac{1}{2\pi\sqrt{LC}}$.

Thus, $$C = \frac{1}{\left(2\pi f_0\right)^2 L} = \frac{1}{\left[2\pi\left(6.30 \times 10^6\ \text{Hz}\right)\right]^2\left(1.05 \times 10^{-6}\ \text{H}\right)} = \boxed{608\ \text{pF}}$$

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P32.49 At different times, $(U_C)_{\max} = (U_L)_{\max}$, so

$$\left[\frac{1}{2}C(\Delta V)^2\right]_{\max} = \frac{1}{2}LI_i^2$$

Then,

$$I_i = \sqrt{\frac{C}{L}}(\Delta V)_{\max} = \sqrt{\frac{1.00\times10^{-6}\ \text{F}}{10.0\times10^{-3}\ \text{H}}}(40.0\ \text{V})$$
$$= 0.400\ \text{A} = \boxed{400\ \text{mA}}$$

P32.50 From the angular frequency of oscillation of the circuit, we have

$$\omega = \frac{1}{\sqrt{LC}} = 2\pi f$$

Solving for the inductance gives

$$L = \frac{1}{C(2\pi f)^2} = \frac{1}{(8.00\times10^{-6}\ \text{F})[2\pi(120\ \text{Hz})]^2}$$
$$= \boxed{0.220\ \text{H}}$$

P32.51 At different times, the maximum energy stored in the capacitor is equal to the maximum energy stored in the inductor.

$$\left[\frac{1}{2}C(\Delta V)^2\right]_{\max} = \frac{1}{2}LI_i^2$$

so

$$(\Delta V_C)_{\max} = \sqrt{\frac{L}{C}}I_i = \sqrt{\frac{20.0\times10^{-3}\ \text{H}}{0.500\times10^{-6}\ \text{F}}}(0.100\ \text{A}) = \boxed{20.0\ \text{V}}$$

P32.52 Find the energy stored in the circuit from Equation 32.27:

$$U = \frac{Q_{\max}^2}{2C} = \frac{(200\times10^{-6}\ \text{C})^2}{2(50.0\times10^{-6}\ \text{F})} = 4.00\times10^{-4}\ \text{J} = 400\ \mu\text{J}$$

If the energy is split equally between the capacitor and inductor at some instant, the energy would be half this value, or 200 μJ. Therefore, there would be no time when each component stores 250 μJ.

P32.53 (a) The frequency of oscillation of the circuit is

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.100\ \text{H})(1.00\times10^{-6}\ \text{F})}} = \boxed{503\ \text{Hz}}$$

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(b) The maximum charge on the capacitor is

$$Q = C\mathcal{E} = \left(1.00\times10^{-6}\ \text{F}\right)(12.0\ \text{V}) = \boxed{12.0\ \mu\text{C}}$$

(c) To find the maximum current I_i, we equate

$$\frac{1}{2}C\mathcal{E}^2 = \frac{1}{2}LI_i^2$$

Then solve for I_i to obtain

$$I_i = \mathcal{E}\sqrt{\frac{C}{L}} = 12.0\ \text{V}\sqrt{\frac{1.00\times10^{-6}\ \text{F}}{0.100\ \text{H}}} = \boxed{37.9\ \text{mA}}$$

(d) The total energy the circuit possesses at t = 3.00 s and at all times is

$$U = \frac{1}{2}C\mathcal{E}^2 = \frac{1}{2}\left(1.00\times10^{-6}\ \text{F}\right)(12.0\ \text{V})^2 = \boxed{72.0\ \mu\text{J}}$$

P32.54 At t = 0 the capacitor charge is at its maximum value, so $\phi = 0$ in

$$Q = Q_{\max}\cos(\omega t + \phi) = Q_{\max}\cos\left(\frac{t}{\sqrt{LC}}\right)$$

Substituting the given information, the charge at 2 ms is

$$\begin{aligned} Q &= (105\times10^{-6}\ \text{C})\cos\left(\frac{2.00\ \times\ 10^{-3}\ \text{s}}{\sqrt{(3.30\ \text{H})(840\times10^{-12}\ \text{F})}}\right) \\ &= (105\times10^{-6}\ \text{C})\cos(38.0\ \text{rad}) \\ &= 1.01\times10^{-4}\ \text{C} \end{aligned}$$

(a) Then the energy in the capacitor is

$$U_C = \frac{Q^2}{2C} = \frac{\left(1.01\ \times\ 10^{-4}\ \text{C}\right)^2}{2\left(840\ \times\ 10^{-12}\ \text{F}\right)} = \boxed{6.03\ \text{J}}$$

(c) The constant total energy is that originally of the capacitor:

$$U = \frac{Q_{\max}^2}{2C} = \frac{\left(1.05\ \times\ 10^{-4}\ \text{C}\right)^2}{2\left(840\ \times\ 10^{-12}\ \text{F}\right)} = \boxed{6.56\ \text{J}}$$

(b) So the inductor's energy is the remaining

$$U_L = 6.56\ \text{J} - 6.03\ \text{J} = \boxed{0.529\ \text{J}}$$

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P32.55 (a) The angular frequency of oscillations is

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.082\,0\ \text{H})(17.0\times10^{-6}\ \text{F})}} = 847\ \text{rad/s} = 2\pi f$$

so $f = \boxed{135\ \text{Hz}}$

(b) The charge on the capacitor is

$$Q = Q_{\max}\cos\omega t = (180\ \mu\text{C})\cos\left[(847\ \text{rad/s})(0.001\,00\ \text{s})\right]$$

$$= \boxed{119\ \mu\text{C}}$$

(c) The current in the circuit is given by Equation 32.23:

$$i = \frac{dq}{dt} = -\omega Q_{\max}\sin\omega t$$

$$= -(847\ \text{rad/s})(180\ \mu\text{C})\sin\left[(847\ \text{Hz})(0.001\,00\ \text{s})\right]$$

$$= \boxed{-114\ \text{mA}}$$

Section 32.6 The *RLC* Circuit

P32.56 We choose to call positive current clockwise in Figure 32.15. It drains charge from the capacitor according to $i = -\frac{dq}{dt}$. A clockwise trip around the circuit then gives

$$+\frac{q}{C} - iR - L\frac{di}{dt} = 0$$

or $$+\frac{q}{C} + \frac{dq}{dt}R + L\frac{d}{dt}\frac{dq}{dt} = 0,$$ identical to Equation 32.28.

P32.57 (a) The frequency of damped oscillations is given by Equation 32.32:

$$\omega_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$= \sqrt{\frac{1}{(2.20\times10^{-3}\,\text{H})(1.80\times10^{-6}\,\text{F})} - \left(\frac{7.60}{2(2.20\times10^{-3}\,\text{H})}\right)^2}$$

$$= 1.58\times10^{4}\ \text{rad/s}$$

Therefore, $f_d = \frac{\omega_d}{2\pi} = \frac{1.58\times10^{4}\ \text{rad/s}}{2\pi} = \boxed{2.51\ \text{kHz}}$.

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(b) Critical damping occurs when $\omega_d = 0$, or when

$$R_c = \sqrt{\frac{4L}{C}} = \sqrt{\frac{4\left(2.20\times10^{-3}\,\text{H}\right)}{1.80\times10^{-6}\,\text{F}}} = \boxed{69.9\ \Omega}$$

P32.58 (a) The angular frequency of undamped oscillations is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.500\ \text{H})\left(0.100\times10^{-6}\ \text{F}\right)}} = 4.47\times10^3\ \text{rad/s}$$

$$= \boxed{4.47\ \text{krad/s}}$$

(b) The frequency of the damped oscillations is

$$\omega_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$= \sqrt{\frac{1}{(0.500\ \text{H})\left(0.100\times10^{-6}\ \text{F}\right)} - \left[\frac{1.00\times10^3\ \Omega}{2(0.500\ \text{H})}\right]^2}$$

$$= \boxed{4.36\ \text{krad/s}}$$

(c) $$\frac{\Delta\omega}{\omega_0}\times100\% = \frac{\omega_d - \omega_0}{\omega_0}\times100\% = \frac{4.36 - 4.47}{4.47}\times100\% = \boxed{-2.53\%}$$

P32.59 (a) The charge on the capacitor is given by Equation 32.31:

$$q = Q_{\max}e^{-Rt/2L}\cos\omega_d t \qquad \text{so} \qquad I_i \propto e^{-Rt/2L}$$

When the amplitude of the oscillation falls to 50.0% of its initial value, we have

$$0.500 = e^{-Rt/2L} \qquad \text{and} \qquad \frac{Rt}{2L} = -\ln(0.500)$$

Then,

$$t = -\frac{2L}{R}\ln(0.500) = \boxed{0.693\left(\frac{2L}{R}\right)}$$

(b) The initial energy of the circuit is $U_0 \propto Q_{\max}^2$. When $U = 0.500U_0$,

$$q = \sqrt{0.500}\,Q_{\max} = 0.707Q_{\max}$$

Then,

$$t = -\frac{2L}{R}\ln(0.707) = \boxed{0.347\left(\frac{2L}{R}\right)}$$ (half as long as part (a))

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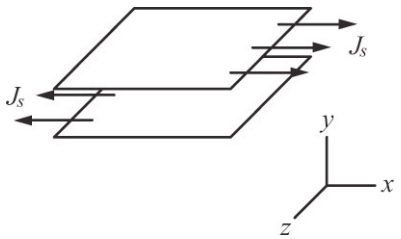
Additional Problems

P32.60 (a) Let Q represent the magnitude of the opposite charges on the plates of a parallel plate capacitor, the two plates having area A and separation d. The negative plate creates an electric field $\vec{\mathbf{E}} = \dfrac{Q}{2\epsilon_0 A}$ toward itself. It exerts on the positive plate force $\vec{\mathbf{F}} = \dfrac{Q^2}{2\epsilon_0 A}$ toward the negative plate. The total field between the plates is $\dfrac{Q}{\epsilon_0 A}$. The energy density is $u_E = \dfrac{1}{2}\epsilon_0 E^2 = \dfrac{1}{2}\epsilon_0 \dfrac{Q^2}{\epsilon_0^2 A^2} = \dfrac{Q^2}{2\epsilon_0 A^2}$. Modeling this as a negative or inward pressure, we have for the force on one plate $F = PA = \dfrac{Q^2}{2\epsilon_0 A}$, in agreement with our first analysis.

(b) The lower of the two current sheets shown creates above it magnetic field $\vec{\mathbf{B}} = \dfrac{\mu_0 J_s}{2}\left(-\hat{\mathbf{k}}\right)$. Let ℓ and w represent the length and width of each sheet. The upper sheet carries current $J_s w$ and feels force

$$\vec{\mathbf{F}} = I\vec{\ell} \times \vec{\mathbf{B}} = J_s w\left[\ell\hat{\mathbf{i}} \times \left(-\frac{\mu_0 J_s}{2}\hat{\mathbf{k}}\right)\right] = \frac{\mu_0 w \ell J_s^2}{2}\hat{\mathbf{j}}$$

The force per area is $P = \dfrac{F}{\ell w} = \boxed{\dfrac{\mu_0 J_s^2}{2}}$.



ANS. FIG. P32.60(b)

(c) Between the two sheets, each sheet contributes the same field, so the total magnetic field is $\dfrac{\mu_0 J_s}{2}\left(-\hat{\mathbf{k}}\right) + \dfrac{\mu_0 J_s}{2}\left(-\hat{\mathbf{k}}\right) = \mu_0 J_s \hat{\mathbf{k}}$, with magnitude $\boxed{B = \mu_0 J_s}$. Outside the space they enclose, the fields of the separate sheets are in opposite directions and add to $\boxed{\text{zero}}$.

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(d) $u_B = \dfrac{1}{2\mu_0}B^2 = \dfrac{\mu_0^2 J_s^2}{2\mu_0} = \boxed{\dfrac{\mu_0 J_s^2}{2}}$

(e) The energy density found in part (d) agrees with the magnetic pressure found in part (b).

P32.61 (a) The voltage across the inductor is given by

$$\mathcal{E}_L = -L\frac{di}{dt} = -(1.00\text{ mH})\frac{d(20.0t)}{dt} = \boxed{-20.0\text{ mV}}$$

(b) The charge that flows into the capacitor is

$$q = \int_0^t I\,dt = \int_0^t (20.0t)dt = 10.0t^2$$

Going across the capacitor in the directon of the current, the potential drops from the positive to the negative side, so

$$\Delta V_C = \frac{-q}{C} = \frac{-10.0t^2}{1.00\times10^{-6}\text{ F}} = \boxed{-10.0t^2}$$

where ΔV_C is in megavolts and t is in seconds.

(c) When $\dfrac{q^2}{2C} \geq \dfrac{1}{2}Li^2$, or

$$\frac{\left(-10.0t^2\right)^2}{2\left(1.00\times10^{-6}\right)} \geq \frac{1}{2}\left(1.00\times10^{-3}\right)(20.0t)^2,$$

then $100t^4 \geq \left(400\times10^{-9}\right)t^2$

The earliest time this is true is at

$$t = \sqrt{4.00\times10^{-9}}\ s = \boxed{63.2\ \mu\text{s}}$$

P32.62 (a) The voltage across the inductor is given by

$$\mathcal{E}_L = -L\frac{di}{dt} = -L\frac{d}{dt}(Kt) = \boxed{-LK}$$

(b) The current into the capacitor is $i = \dfrac{dq}{dt}$, so the charge that flows into the capacitor is

$$q = \int_0^t i\,dt = \int_0^t Kt\,dt = \frac{1}{2}Kt^2$$

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Going across the capacitor in the direction of the current, the potential drops from the positive to the negative side, so

$$\Delta V_C = \frac{-q}{C} = \boxed{-\frac{Kt^2}{2C}}$$

(c) When $\frac{1}{2}C(\Delta V_C)^2 \geq \frac{1}{2}LI^2$,

$$\frac{1}{2}C\left(\frac{K^2t^4}{4C^2}\right) \geq \frac{1}{2}L\left(K^2t^2\right)$$

Thus, the energy in the capacitor begins to exceed the energy in the inductor after $t = \boxed{2\sqrt{LC}}$.

P32.63 The total energy equals the energy in the capacitor and inductor:

$$\frac{1}{2}\frac{Q^2}{C} = \frac{1}{2C}\left(\frac{Q}{2}\right)^2 + \frac{1}{2}Li^2$$

so $$i = \sqrt{\frac{3Q^2}{4CL}}$$

The flux through each turn of the coil is

$$\Phi_B = \frac{Li}{N} = \boxed{\frac{q}{2N}\sqrt{\frac{3L}{C}}}$$

where N is the number of turns.

P32.64 (a) The inductor has no resistance, therefore it has voltage across it. It behaves as a $\boxed{\text{short circuit}}$.

(b) The battery sees an equivalent resistance

$$4.00\ \Omega + \left(\frac{1}{4.00\ \Omega} + \frac{1}{8.00\ \Omega}\right)^{-1} = 6.67\ \Omega$$

The battery current is

$$\frac{10.0\ \text{V}}{6.67\ \Omega} = 1.50\ \text{A}$$

The voltage across the parallel combination of resistors is

$$10.0\ \text{V} - (1.50\ \text{A})(4.00\ \Omega) = 4.00\ \text{V}$$

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The current in the 8-Ω resistor and the inductor is

$$\frac{4.00\text{ V}}{8.00\ \Omega} = \boxed{500\text{ mA}}$$

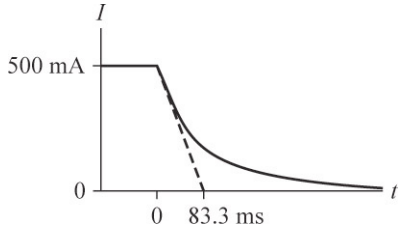
(c) The energy stored in the inductor for $t < 0$ is

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}(1.00\text{ H})(0.500\text{ A})^2 = \boxed{125\text{ mJ}}$$

(d) The energy becomes 125 mJ of additional internal energy in the 8-Ω resistor and the 4-Ω resistor in the middle branch.

(e) See ANS. FIG. P32.64(e). The current decreases from 500 mA toward zero, showing exponential decay with a time constant

$$\tau = \frac{L}{R} = \frac{1.00\text{ H}}{3(4.00\ \Omega)} = 0.083\ 3\text{ s} = 8.33\text{ ms}$$



ANS. FIG. P32.64(e)

P32.65 The voltages are related as

$$\mathcal{E} - iR - L\frac{di}{dt} = 0 \quad \rightarrow \quad \mathcal{E} = iR + L\frac{di}{dt} = 0$$

When the current is increasing:

$$9.00\text{ V} = (2.00\text{ A})R + L(0.500\text{ A/s}) \qquad \textbf{[1]}$$

When the current is decreasing:

$$5.00\text{ V} = (2.00\text{ A})R + L(-0.500\text{ A/s}) \qquad \textbf{[2]}$$

(a) Subtracting [2] from [1] gives

$$9.00\text{ V} - 5.00\text{ V} = L(1.00\text{ A/s}) \quad \rightarrow \quad \boxed{L = 4.00\text{ H}}$$

(b) Substituting the value for L in either equation gives

$$7.00\text{ V} = (2.00\text{ A})R \quad \rightarrow \quad \boxed{R = 3.50\ \Omega}$$

P32.66 (a) Initially, the current is zero because of the emf induced in the coil resists an increase in the current. Just after the circuit is connected, the potential difference across the resistor is 0 and the emf across the coil is 24.0 V.

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(b) After several seconds, the current has reached a steady value and does not change. After several seconds, the potential difference across the resistor is 24.0 V and that across the coil is 0.

(c) The resistor voltage and inductor voltage always add to 24 V. The resistor voltage increases monotonically, so the two voltages are equal to each other, both being 12.0 V, just once. The time is given by

$$V = iR = R\mathcal{E}/R(1-e^{-Rt/L})$$

Substituting,

$$12\text{ V} = 24\text{ V}(1 - e^{-6\Omega t/0.005\text{ H}})$$

$$0.5 = e^{-1\ 200\ t} \rightarrow 1\ 200\ t = \ln\ 2 \quad \rightarrow \quad t = 0.578\text{ ms}$$

The two voltages are equal to each other, both being 12.0 V, just once, at 0.578 ms after the circuit is connected.

(d) There is now no battery in the circuit, so the current decays to zero. The resistor and inductor are in parallel because they have common connections on each side. As the current decays the potential difference across the resistor is always equal to the emf across the coil.

P32.67 (a) At the center, $B \approx \dfrac{N\mu_0 i}{2R}$.

So the coil creates flux through itself

$$\Phi_B = BA\cos\theta = \frac{N\mu_0 i}{2R}\pi R^2 \cos 0° = \frac{1}{2}N\mu_0 \pi i R$$

The inductance is

$$L = N\frac{\Phi_B}{i} \approx N\left(\frac{N\mu_0 \pi I R}{2i}\right) \approx \boxed{\frac{1}{2}\mu_0 \pi N^2 R}$$

(b) To find the inductance of the circuit, we compute its radius from

$$2\pi R = 3(0.300\text{ m}) \quad \rightarrow \quad R = 0.143\text{ m}$$

Then, from the expression found in part(a), the inductance is

$$L \approx \frac{1}{2}\mu_0 \pi N^2 R = \frac{1}{2}\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\pi\left(1^2\right)(0.143\text{ m})$$

$$= 2.83 \times 10^{-7}\ \text{H}$$

$$\boxed{L \sim 10^{-7}\ \text{H}}$$

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(c) The time constant is

$$\tau = \frac{L}{R} \approx \frac{2.83\times10^{-7}\ \text{H}}{270\ \Omega} = 1.05\times10^{-9}\ \text{s} \approx \boxed{10^{-9}\ \text{s}} \qquad \boxed{\frac{L}{R} \sim 1\ \text{ns}}$$

P32.68 We calculate the angular frequency of the circuit from Equation 32.32:

$$\omega_d = \left[\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right]^{1/2}$$

$$= \left[\frac{1}{(32.0\times10^{-3}\ \text{H})(500\times10^{-6}\ \text{F})} - \left(\frac{16.0\ \Omega}{2(32.0\times10^{-3}\ \text{H})}\right)^2\right]^{1/2}$$

$$= 0$$

The fact that the angular frequency at which the circuit oscillates is zero tells you that the circuit is critically damped. There will be no decaying oscillations. The critical resistance is given by

$$R_c = \sqrt{\frac{4L}{C}} = \sqrt{\frac{4(32.0\times10^{-3}\ \text{H})}{500\times10^{-6}\ \text{F}}} = 16.0\ \Omega$$

which is just the resistance that you are using for your experiment.

P32.69 The emf across the inductor is given by

$$\mathcal{E} = -L\frac{di}{dt} = -L\frac{\Delta i}{\Delta t} = -50\frac{\Delta i}{\Delta t}$$

where the rate of change of current $\frac{\Delta i}{\Delta t}$ is in amperes per second (A/s), and the induced emf $\mathcal{E}$ is in millivolts (mV).

Between $t = 0$ and $t = 1$ ms: $\frac{\Delta i}{\Delta t} = 2$ $\mathcal{E} = -100$ mV

Between $t = 1$ ms and $t = 2$ ms: $\frac{\Delta i}{\Delta t} = 0$ $\mathcal{E} = 0$

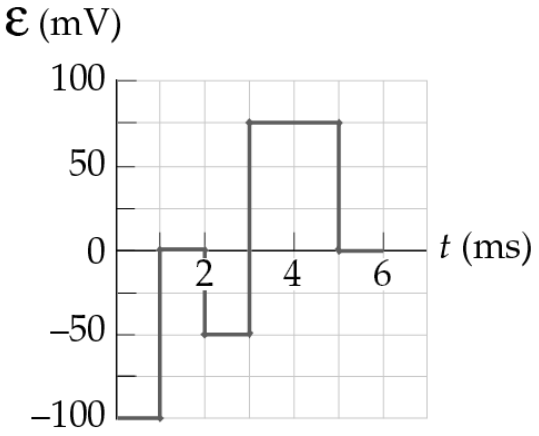
Between $t = 2$ ms and $t = 3$ ms: $\frac{\Delta i}{\Delta t} = 1$ $\mathcal{E} = -50$ mV

Between $t = 3$ ms and $t = 5$ ms: $\frac{\Delta i}{\Delta t} = -\frac{3}{2}$ $\mathcal{E} = +75$ mV

Between $t = 5$ ms and $t = 6$ ms: $\frac{\Delta i}{\Delta t} = 0$ $\mathcal{E} = 0$

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The graph of $\mathcal{E}$ is shown in ANS. FIG. P32.69.



ANS. FIG. P32.69

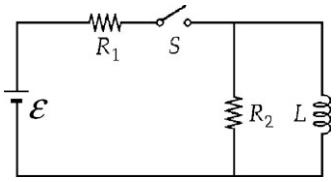
P32.70 (a) $\boxed{i_1 = i_2 + i}$

(b) For the left-hand loop,

$$\boxed{\mathcal{E} - i_1 R_1 - i_2 R_2 = 0}$$

(c) For the outside loop,

$$\boxed{\mathcal{E} - i_1 R_1 - L\frac{di}{dt} = 0}$$



ANS. FIG. P32.70

(d) Substitute the equation for i_1 from part (a) into the equation in part (b):

$$\mathcal{E} - (i_2 + i)R_1 - i_2 R_2 = 0 \quad \rightarrow \quad i_2 = \frac{\mathcal{E} - iR_1}{R_1 + R_2}$$

Substitute the equation for i_1 from part (a) into the equation in part (c):

$$\mathcal{E} - (i_2 + i)R_1 - L\frac{di}{dt} = 0 \quad \rightarrow \quad i_2 = \frac{\mathcal{E} - L\dfrac{di}{dt}}{R_1} - i$$

Equate the two expressions for i_2:

$$\frac{\mathcal{E} - iR_1}{R_1 + R_2} = \frac{\mathcal{E} - L\dfrac{di}{dt}}{R_1} - i$$

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Expanding and solving,

$$\mathcal{E} - L\frac{di}{dt} = \left(\frac{\mathcal{E} - iR_1}{R_1 + R_2} + i\right)R_1$$

$$= \left[\frac{\mathcal{E} - iR_1 + i(R_1 + R_2)}{R_1 + R_2}\right]R_1 = \left(\frac{\mathcal{E} + iR_2}{R_1 + R_2}\right)R_1$$

$$L\frac{di}{dt} = \mathcal{E} - \left(\frac{\mathcal{E} + iR_2}{R_1 + R_2}\right)R_1$$

$$= \frac{\mathcal{E}(R_1 + R_2) - (\mathcal{E} + iR_2)R_1}{R_1 + R_2} = \frac{\mathcal{E}(R_2) - (iR_2)R_1}{R_1 + R_2}$$

And finally,

$$\mathcal{E}\frac{R_2}{R_1 + R_2} - i\frac{R_1R_2}{R_1 + R_2} - L\frac{di}{dt} = 0$$

Calling $\mathcal{E}' = \mathcal{E}\dfrac{R_2}{R_1 + R_2}$ and $R' = \dfrac{R_1R_2}{R_1 + R_2}$, the equation for i can be written

$$\boxed{\mathcal{E}' - iR' - L\frac{di}{dt} = 0}$$

(e) This is of the same form as the Equation 32.6 in the text for a simple *RL* circuit, so its solution is of the same form as Equation 32.7:

$$i = \frac{\mathcal{E}'}{R'}\left(1 - e^{-R't/L}\right)$$

where

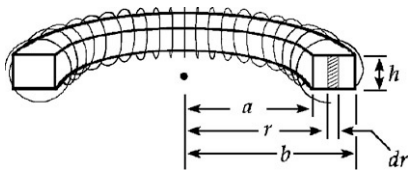
$$\frac{\mathcal{E}'}{R'} = \frac{\mathcal{E}R_2/(R_1 + R_2)}{R_1R_2/(R_1 + R_2)} = \frac{\mathcal{E}}{R_1}$$

Thus,

$$i = \frac{\mathcal{E}}{R_1}\left(1 - e^{-R't/L}\right) \quad \text{where} \quad R' = \frac{R_1R_2}{R_1 + R_2}$$

P32.71 See ANS. FIG. 32.71. The magnetic field inside the toroid is given by

$$B = \frac{\mu_0 Ni}{2\pi r}$$



ANS. FIG. P32.71

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The magnetic flux through a cross-sectional area $A = h(b - a)$ is given by

$$\Phi_B = \int B\,dA = \int_a^b \frac{\mu_0 Ni}{2\pi r} h\,dr = \frac{\mu_0 Nih}{2\pi}\int_a^b \frac{dr}{r} = \frac{\mu_0 Nih}{2\pi}\ln\left(\frac{b}{a}\right)$$

Thus, the inductance is

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi}\ln\left(\frac{b}{a}\right)$$

Substituting numerical values, we obtain

$$L = \frac{\mu_0 (500)^2 (0.010\ 0\ \text{m})}{2\pi}\ln\left(\frac{12.0\ \text{cm}}{10.0\ \text{cm}}\right) = \boxed{91.2\ \mu\text{H}}$$

P32.72 See ANS. FIG. P23.71. $B = \dfrac{\mu_0 Ni}{2\pi r}$ inside the toroid.

Calculate the flux through a cross-sectional area $A = h(b - a)$:

$$\Phi_B = \int B\,dA = \int_a^b \frac{\mu_0 Ni}{2\pi r} h\,dr = \frac{\mu_0 Nih}{2\pi}\int_a^b \frac{dr}{r} = \frac{\mu_0 Nih}{2\pi}\ln\left(\frac{b}{a}\right)$$

Thus, the inductance is

$$L = \frac{N\Phi_B}{i} = \boxed{\frac{\mu_0 N^2 h}{2\pi}\ln\left(\frac{b}{a}\right)}$$

P32.73 (a) $U_B = \dfrac{1}{2}Li^2 = \dfrac{1}{2}(50.0\ \text{H})\left(50.0\times 10^3\ \text{A}\right)^2 = \boxed{6.25\times 10^{10}\ \text{J}}$

(b) Two adjacent turns are parallel wires carrying current in the same direction. Since the loops have such large radius, a one-meter section can be regarded as straight.

Then one wire creates a field of $B = \dfrac{\mu_0 i}{2\pi r}$.

This causes a force on the next wire of $F = i\ell B\sin\theta$,

giving a force per unit length $\dfrac{F}{\ell} = i\dfrac{\mu_0 i}{2\pi r}\sin 90° = \dfrac{\mu_0 i^2}{2\pi r}$.

Evaluating,

$$\frac{F}{\ell} = \left(4\pi\times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\frac{\left(50.0\times 10^3\ \text{A}\right)^2}{2\pi(0.250\ \text{m})} = \boxed{2\ 000\ \text{N/m}}$$

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P32.74 For an *RL* circuit, $i = I_i e^{-(R/L)t}$ and

$$\frac{i}{I_i} = 1 - 10^{-9} = e^{-(R/L)t} \cong 1 - \frac{R}{L}t$$

$$\frac{R}{L}t = 10^{-9}$$

so

$$R_{\max} = \frac{L\left(10^{-9}\right)}{t} = \frac{\left(3.14 \times 10^{-8}\ \text{H}\right)\left(10^{-9}\right)}{\left(2.50\ \text{yr}\right)}\left(\frac{1\ \text{yr}}{3.16 \times 10^{7}\ \text{s}}\right)$$

$$= \boxed{3.97 \times 10^{-25}\ \Omega}$$

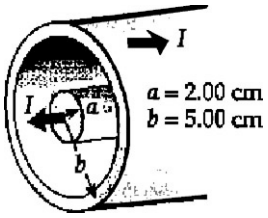
(If the ring were of purest copper, of diameter 1 cm, and cross-sectional area 1 mm^2, its resistance would be at least $10^{-6}\ \Omega$.)

P32.75 Find the current in the cylinder.

$$P = i\Delta V \quad \rightarrow \quad i = \frac{P}{\Delta V} = \frac{1.00 \times 10^{9}\ \text{W}}{200 \times 10^{3}\ \text{V}} = 5.00 \times 10^{3}\ \text{A}$$

From Ampère's law,

$$B\left(2\pi r\right) = \mu_0 i_{\text{enclosed}} \quad \text{or} \quad B = \frac{\mu_0 i_{\text{enclosed}}}{2\pi r}$$



ANS. FIG. P32.75

(a) At $r = a = 0.020\ 0$ m, $i_{\text{enclosed}} = 5.00 \times 10^{3}$ A and

$$B = \frac{\left(4\pi \times 10^{-7}\ \text{T} \cdot \text{m/A}\right)\left(5.00 \times 10^{3}\ \text{A}\right)}{2\pi\left(0.020\ 0\ \text{m}\right)} = 0.050\ 0\ \text{T} = \boxed{50.0\ \text{mT}}$$

(b) At $r = b = 0.050\ 0$ m, $i_{\text{enclosed}} = i = 5.00 \times 10^{3}$ A and

$$B = \frac{\left(4\pi \times 10^{-7}\ \text{T} \cdot \text{m/A}\right)\left(5.00 \times 10^{3}\ \text{A}\right)}{2\pi\left(0.050\ 0\ \text{m}\right)} = 0.020\ 0\ \text{T} = \boxed{20.0\ \text{mT}}$$

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(c) The energy density is $u_B = \dfrac{B^2}{2\mu_0}$:

$$U_B = \int u\,dV = \int_a^b \frac{[B(r)]^2 (2\pi r \ell\, dr)}{2\mu_0} = \frac{\mu_0 i^2 \ell}{4\pi}\int_a^b \frac{dr}{r} = \frac{\mu_0 i^2 \ell}{4\pi}\ln\left(\frac{b}{a}\right)$$

$$U_B = \frac{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(5.00\times10^{3}\ \text{A}\right)^2\left(1\,000\times10^{3}\ \text{m}\right)}{4\pi} \times\ln\left(\frac{5.00\ \text{cm}}{2.00\ \text{cm}}\right)$$

$$= 2.29\times10^{6}\ \text{J} = \boxed{2.29\ \text{MJ}}$$

(d) The magnetic field created by the inner conductor exerts a force of repulsion on the current in the outer sheath. The strength of this field, from part (b), is 20.0 mT. Consider a small rectangular section of the outer cylinder of length ℓ and width w.

It carries a current of $\left(5.00\times10^{3}\ \text{A}\right)\left(\dfrac{w}{2\pi\left(0.050\,0\ \text{m}\right)}\right)$

and experiences an outward force

$$F = i\ell B\sin\theta = \frac{\left(5.00\times10^{3}\ \text{A}\right)w}{2\pi\left(0.050\,0\ \text{m}\right)}\ell\left(20.0\times10^{-3}\ \text{T}\right)\sin 90.0°$$

The pressure on it is

$$P = \frac{F}{A} = \frac{F}{w\ell} = \frac{\left(5.00\times10^{3}\ \text{A}\right)\left(20.0\times10^{-3}\ \text{T}\right)\cancel{w\ell}}{2\pi\left(0.050\,0\ \text{m}\right)\cancel{w\ell}} = \boxed{318\ \text{Pa}}$$

P32.76 (a) The magnetic field inside the solenoid is given by $B = \dfrac{\mu_0 N i}{\ell}$:

$$B = \frac{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)(1\,400)(2.00\ \text{A})}{1.20\ \text{m}}$$

$$= 2.93\times10^{-3}\ \text{T} = \boxed{2.93\ \text{mT}}$$



ANS. FIG. P32.76

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(b) The energy density of the magnetic field is

$$u_B = \frac{B^2}{2\mu_0} = \frac{\left(2.93\times10^{-3}\ \text{T}\right)^2}{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)} = \left(3.42\ \text{J/m}^3\right)\left(\frac{1\ \text{N}\cdot\text{m}}{1\ \text{J}}\right)$$
$$= 3.42\ \text{N/m}^2 = \boxed{3.42\ \text{Pa}}$$

(c) The supercurrents must be clockwise to produce a downward magnetic field to cancel the upward field of the current in the windings.

(d) The field of the windings is upward and radially outward around the top of the solenoid. It exerts a force radially inward and upward on each bit of the clockwise supercurrent. The total force on the supercurrents in the bar is upward. You can think of it as a force of repulsion between the solenoid with its north end pointing up, and the core, with its north end pointing down.

(e) $F = PA = \left(3.42\ \text{Pa}\right)\left[\pi\left(1.10\times10^{-2}\ \text{m}\right)^2\right] = 1.30\times10^{-3}\ \text{N} = \boxed{1.30\ \text{mN}}$

Note that we have not proved that energy density is pressure. In fact, it is not in some cases. Chapter 21 proved that the pressure is two-thirds of the translational energy density in an ideal gas.

P32.77 From Ampère's law, the magnetic field at distance $r \le R$ is found as:

$$B\left(2\pi r\right) = \mu_0 j\left(\pi r^2\right) = \mu_0\left(\frac{i}{\pi R^2}\right)\left(\pi r^2\right),, \quad \text{or} \quad B = \frac{\mu_0 i r}{2\pi R^2}$$

The magnetic energy per unit length within the wire is then

$$\frac{U_B}{\ell} = \int_0^R \frac{B^2}{2\mu_0}\left(2\pi r dr\right) = \frac{\mu_0 i^2}{4\pi R^4}\int_0^R r^3\, dr = \frac{\mu_0 i^2}{4\pi R^4}\left(\frac{R^4}{4}\right) = \boxed{\frac{\mu_0 i^2}{16\pi}}$$

This is independent of the radius of the wire.

Challenge Problems

P32.78 (a) It has a magnetic field, and it stores energy, so $L = \dfrac{2U_B}{i^2}$ is non-zero.

(b) Every field line goes through the rectangle between the conductors.

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(c) When the wires carry current i, magnetic flux passes through the rectangle bordered by the wires (surface to surface of the wires):

$$L = \frac{\Phi}{i} = \frac{1}{i}\int_{y=a}^{w-a} B\,dA$$

where y is measured from the center of the lower wire, dA is a rectangular area element of length x and width dy, and B is the magnitude of the net magnetic field generated by the upper and lower wires that passes through dA. The inductance is

$$L = \frac{1}{i}\int_{a}^{w-a}\left[\frac{\mu_0 i}{2\pi y} + \frac{\mu_0 i}{2\pi(w-y)}\right] x\,dy$$

We can simplify this calculation by noting that by the symmetry of the arrangement, each conductor contributes equally to the field that passes through the area between them. Thus, the total inductance of both wires is twice the inductance of one wire. The inductance due to the lower wire is

$$L_{\text{lower}} = \frac{1}{i}\int_{a}^{w-a}\frac{\mu_0 i}{2\pi y}x\,dy = \frac{\mu_0 x}{2\pi}\ln y\Big|_{a}^{w-a} = \frac{\mu_0 x}{2\pi}\left[\ln(w-a) - \ln a\right]$$

$$= \frac{\mu_0 x}{2\pi}\ln\left(\frac{w-a}{a}\right)$$

The inductance due to both wires is twice this: $L = \dfrac{\mu_0 x}{\pi}\ln\left(\dfrac{w-a}{a}\right)$.

P32.79 The total magnetic energy is the volume integral of the energy density,

$$u_B = \frac{B^2}{2\mu_0}$$

Because B changes with position, u_B is not constant. For $B = B_0\left(\dfrac{R}{r}\right)^2$,

$$u_B = \left(\frac{B_0^2}{2\mu_0}\right)\left(\frac{R}{r}\right)^4$$

Next, we set up an expression for the magnetic energy in a spherical shell of radius r and thickness dr. Such a shell has a volume $4\pi r^2 dr$, so the energy stored in it is

$$dU_B = u_B\left(4\pi r^2 dr\right) = \left(\frac{2\pi B_0^2 R^4}{\mu_0}\right)\frac{dr}{r^2}$$

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We integrate this expression for $r = R$ to $r = \infty$ to obtain the total magnetic energy outside the sphere. This gives

$$U_B = \boxed{\frac{2\pi B_0^2 R^3}{\mu_0}}$$

Substituting numerical values,

$$U_B = \frac{2\pi B_0^2 R^3}{\mu_0} = \frac{2\pi\left(5.00\times10^{-5}\ \text{T}\right)^2\left(6.00\times10^{6}\ \text{m}\right)^3}{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)}$$

$$= \boxed{2.70\times10^{18}\ \text{J}}$$

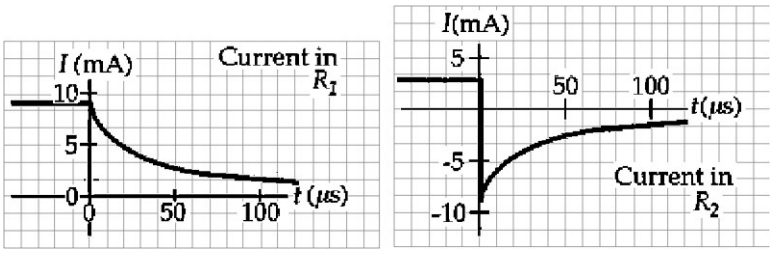
P32.80 (a) While steady-state conditions exist, a 9.00 mA flows clockwise around the right loop of the circuit. Immediately after the switch is opened, a 9.00 mA current will flow around the outer loop of the circuit. Applying Kirchhoff's loop rule going clockwise around this loop gives:

$$+\mathcal{E} - \left[(2.00+6.00)\times10^{3}\ \Omega\right]\left(9.00\times10^{-3}\ \text{A}\right) = 0$$

$$\mathcal{E} = \boxed{72.0\ \text{V}}$$

(b) Starting at point *a*, the potential rises across the inductor then falls across resistors R_2 and R_1. The positive answer in part (a) means that $\boxed{\text{point b}}$ is the higher potential.

(c) The currents in R_1 and R_2 are shown in ANS. FIG. P32.80(c).below. After $t = 0$, the current in R_1 decreases from an initial value of 9.00 mA according to $i = I_i e^{-Rt/L}$. Taking the original current direction as positive in each resistor, the current decreases from +9.00 mA (to the right) to zero in R_1. In R_2 the current jumps from +3.00 mA (downward) to –9.00 mA (upward) and then decreases in magnitude to zero. The time constant of each decay is 0.4 H/8 000 Ω = 50 μs. Thus we draw each current dropping to $1/e$ = 36.8% of its original value = 3.3 μA at the 50 μs instant.



ANS. FIG. P32.80(c)

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(d) After the switch is opened, the current around the outer loop decays as

$$i = I_i e^{-Rt/L}$$

with $I_i = 9.00$ mA, $R = 8.00$ kΩ, and $L = 0.400$ H.

Thus, when the current has reached a value $i = 2.00$ mA, the elapsed time is:

$$t = \left(\frac{L}{R}\right)\ln\left(\frac{I_i}{i}\right) = \left(\frac{0.400\text{ H}}{8.00\times10^3\ \Omega}\right)\ln\left(\frac{9.00\text{ mA}}{2.00\text{ mA}}\right)$$

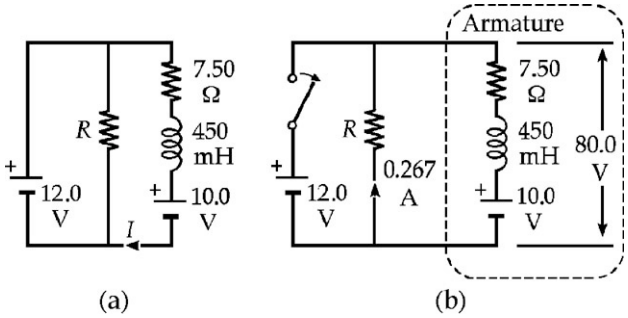
$$= 7.52\times10^{-5}\text{ s} = \boxed{75.2\ \mu\text{s}}$$

P32.81 When the switch is closed, as shown in ANS. FIG. P32.81(a), the current in the inductor is i:

$$12.0 - 7.50i - 10.0 = 0 \rightarrow i = 0.267\text{ A}$$

When the switch is opened, as shown in ANS. FIG. P32.81(b), the initial current in the inductor remains at 0.267 A. Across resistor R and the armature, $iR = \Delta V$ and

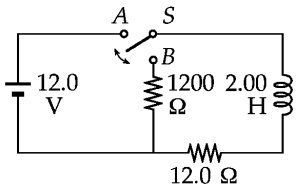
$$(0.267\text{ A})R \le 80.0\text{ V} \quad \rightarrow \quad \boxed{R \le 300\ \Omega}$$



ANS. FIG. P32.81

***P32.82** (a) After a long time,

$$i = \frac{\mathcal{E}}{R} = \frac{12.0\text{ V}}{12.0\ \Omega} = \boxed{1.00\text{ A}}$$



ANS. FIG. P32.82

(b) With the switch thrown to position b, the emf is no longer part of the circuit. The initial current is 1.00 A:

$$\Delta V_{12} = (1.00\text{ A})(12.00\ \Omega) = \boxed{12.0\text{ V}}$$

$$\Delta V_{1\,200} = (1.00\text{ A})(1\,200\ \Omega) = \boxed{1.20\text{ kV}}$$

$$\Delta V_L = \Delta V_R = 1\ 200\text{ V} + 12.0\text{ V} = \boxed{1.21\text{ kV}}$$

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(c) With the switch thrown to position b, $\mathcal{E} = 0$, so

$$\Delta V_L = \Delta V_R = iR_{eff} = i(1\,200\ \Omega + 12.0\ \Omega) = i(1.21\ \text{k}\Omega)$$

Then, when the voltage across the inductor has reached 12.0 V,

$$i = \frac{\Delta V_L}{R_{eff}} = \frac{12.0\ \text{V}}{1.21\ \text{k}\Omega} = 9.90 \times 10^{-3}\ \text{A}$$

The current in the inductor decays as $i = I_i e^{-Rt/L}$. Solving for the time t,

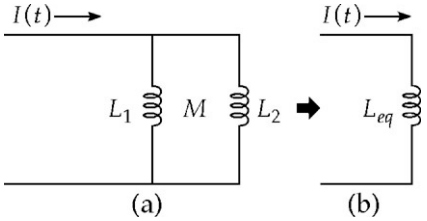
$$t = \frac{L}{R}\ln\left(\frac{I_i}{i}\right) = \left(\frac{2.00\ \text{H}}{1.21\ \text{k}\Omega}\right)\ln\left(\frac{1.00\ \text{A}}{9.90 \times 10^{-3}\ \text{A}}\right) = \boxed{7.62 \times 10^{-3}\ \text{s}}$$

P32.83 With $i = i_1 + i_2$, the voltage across the pair is:

$$\Delta V = -L_1\frac{di_1}{dt} - M\frac{di_2}{dt} = -L_{\text{eq}}\frac{di}{dt} \qquad \textbf{[1]}$$

and

$$\Delta V = -L_2\frac{di_2}{dt} - M\frac{di_1}{dt} = -L_{\text{eq}}\frac{di}{dt} \qquad \textbf{[2]}$$



ANS. FIG. P32.83

So, from [1], we have

$$-\frac{di_1}{dt} = \frac{\Delta V}{L_1} + \frac{M}{L_1}\frac{di_2}{dt}$$

which, when substituted into [2], gives

$$-L_2\frac{di_2}{dt} + M\left(\frac{(\Delta V)}{L_1} + \frac{M}{L_1}\frac{di_2}{dt}\right) = \Delta V$$

$$\left(-L_1L_2 + M^2\right)\frac{di_2}{dt} = \Delta V\left(L_1 - M\right) \qquad \textbf{[3]}$$

From [2], $-\dfrac{di_2}{dt} = \dfrac{\Delta V}{L_2} + \dfrac{M}{L_2}\dfrac{di_1}{dt}$,

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which, when substituted into [1], gives

$$-L_1\frac{di_1}{dt}+M\left(\frac{\Delta V}{L_2}+\frac{M}{L_2}\frac{di_1}{dt}\right)=\Delta V$$

$$\left(-L_1L_2+M^2\right)\frac{di_1}{dt}=\Delta V\left(L_2-M\right) \qquad \textbf{[4]}$$

Adding [3] to [4], we have

$$\left(-L_1L_2+M^2\right)\frac{di}{dt}=\Delta V\left(L_1+L_2-2M\right)$$

So, $$L_{eq}=-\frac{\Delta V}{di/dt}=\boxed{\frac{L_1L_2-M^2}{L_1+L_2-2M}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P32.2 1.36 μH

P32.4 (a) 1.97 mH; (b) 38.0 mA/s

P32.6 (a) 188 μT; (b) $3.33\times10^{-8}\ \text{T}\cdot\text{m}^2$; (c) 0.375 mH; (d) B and Φ_B are proportional to current: L is independent of current.

P32.8 (a) 5.90 mH; (b) 23.6 mV

P32.10 9.77 mm

P32.12 See P32.12 for full explanation.

P32.14 See ANS. FIG. P32.14.

P32.16 (a) 0.139 s; (b) 0.461 s

P32.18 See ANS. FIGs. P32.18(a), (b), and (c).

P32.20 7.67 mH

P32.22 See P32.22 for full explanation.

P32.24 (a) 2.00 ms; (b) 0.176 A; (c) 1.50 A; (d) 3.22 ms

P32.26 (a) $\dfrac{\mathcal{E}}{5R}\left(1-e^{-5Rt/2L}\right)$; (b) $\dfrac{\mathcal{E}}{10R}\left(6-e^{-5Rt/2L}\right)$

P32.28 For $t\le 0$, the current in the inductor is zero; for $0\le t\le 200\ \mu$s, $i_L=(10.0\ \text{A})\left(1-e^{-10\,000t/\text{s}}\right)$; for $t\ge 200\ \mu$s,, $(63.9\ \text{A})e^{-10\,000t/\text{s}}$

P32.30 (a) See P32.30(a) for full explanation; (b) See P32.30(b) for full explanation; (c) See P32.30(c) for full explanation; (d) Yes. See P32.30(d) for full explanation.

P32.32 64.8 mJ

P32.34 (a) 20.0 W; (b) 20.0 W; (c) 0; (d) 20.0 J

P32.36 $\dfrac{L}{2R}$

P32.38 (a) 66.0 W; (b) 45.0 W; (c) 21.0 W; (d) The power supplied by the battery is equal to the sum of the power delivered to the internal resistance of the coil and the power stored in the magnetic field; (e) Yes; (f) Just after t = 0, the current is very small, so the power delivered to the internal resistance of the coil (iR^2) is nearly zero, but the rate of the change of the current is large, so the power delivered to the magnetic field (Ldi/dt) is large, and nearly all the battery power is being stored in the magnetic field. Long after the connection is made, the current is not changing, so no power is being stored in the

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magnetic field, and all the battery power is being delivered to the internal resistance of the coil.

P32.40 80.0 mH

P32.42 1.73 mH

P32.44 (a) $M_{12} = \mu_0 \pi R_2^2 N_1 N_2 / \ell$; (b) $M_{12} = \mu_0 \pi R_2^2 N_1 N_2 / \ell$; (c) They are the same.

P32.46 (a) See P32.46(a) for full explanation; (b) 3.95 nH

P32.48 608 pF

P32.50 0.220 H

P32.52 If the energy is split equally between the capacitor and inductor at some instant, the energy would be half this value, or 200 μJ. Therefore, there would be no time when each component stores 250 μJ.

P32.54 (a) 6.03 J; (b) 0.529 J; (c) 6.56 J

P32.56 See P32.56 for full explanation.

P32.58 (a) 4.47 krad/s; (b) 4.36 krad/s; (c) –2.53%

P32.60 (a) See P32.60(a) for full explanation; (b) $\frac{\mu_0 J_s^2}{2}$; (c) $B = \mu_0 J_s$, zero; (d) $\frac{\mu_0 J_s^2}{2}$; (e) The energy density found in part (d) agrees with the magnetic pressure found in part (b).

P32.62 (a) $-LK$; (b) $-\frac{Kt^2}{2C}$; (c) $2\sqrt{LC}$

P32.64 (a) short circuit; (b) 500 mA; (c) 125 mJ; (d) The energy becomes 125 mJ of additional internal energy in the 8-Ω resistor and the 4-Ω resistor in the middle branch; (e) See ANS FIG P32.64(e).

P32.66 (a) Just after the circuit is connected, the potential difference across the register is 0, and the emf across the coil is 24.0 V; (b) After several seconds, the potential difference across the resistor is 24.0 V and that across the coil is 0; (c) The two voltages are equal to each other, both being 12.0 V, just once, at 0.578 ms after the circuit is connected; (d) As the current decays, the potential difference across the resistor is always equal to the emf across the coil.

P32.68 See P32.68 for full explanation.

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P32.70 (a) $i_1 = i_2 + i$; (b) $\mathcal{E} - i_1 R_1 - i_2 R_2 = 0$; (c) $\mathcal{E} - i_1 R_1 - L\dfrac{di}{dt} = 0$;

(d) $\mathcal{E}' - iR' - L\dfrac{di}{dt} = 0$; (e) See P32.70(e) for full explanation.

P32.72 $\dfrac{\mu_0 N^2 h}{2\pi} \ln\left(\dfrac{b}{a}\right)$

P32.74 $3.97 \times 10^{-25}\ \Omega$

P32.76 (a) 2.93 mT; (b) 3.42 Pa; (c) The supercurrents must be clockwise to produce a downward magnetic field to cancel the upward field of the current in the windings; (d) The field of the windings is upward and radially outward around the top of the solenoid. It exerts a force radially inward and upward on each bit of the clockwise supercurrent. The total force on the supercurrents in the bar is upward; (e) 1.30 mN

P32.78 (a) It has a magnetic field, and it stores energy, so $L = \dfrac{2U_B}{i^2}$ is non-zero;

(b) Every field line goes through the rectangle between the conductors; (c) See P32.78(c) for full explanation.

P32.80 (a) 72.0 V; (b) point b; (c) See ANS. FIG. P32.80(c); (d) 75.2 μs

P32.82 (a) 1.00 A; (b) $\Delta V_{12} = 12.0$ V, $\Delta V_{1200} = 1.20$ kV, $\Delta V_L = 1.21$ kV

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